L01 — study notes

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Lecture 1 — Introduction and accounting review · Map of the notes

PDF. L01-notes.pdf — this whole unit in one printable file (rendered from these notes; rebuild with `python -I ~/.claude/commands/templates/notes_pdf.py Notas/L01` after any edit).

One claim. A firm sells goods and sells promises, and its balance sheet records the split of one stream of cash into senior promises (liabilities) and a junior residual (equity). **Because the split is exhaustive,** $A = L + E$ **holds in market value, not only in the books**, and that additivity is the seed of Modigliani–Miller. **The books measure that split at historical cost and on accruals**, so book equity can be negative while the market values the firm at 122.6 bn (Starbucks), and profit can be positive while cash runs out. **Ratios mix operations with financing**: DuPont shows ROE rising on leverage alone, and ROIC removes it, with $\text{ROE} = \text{ROIC} + \frac{D}{E}[\text{ROIC} - r_D(1-\tau)]$. **Free cash flow** is the number valuation will use: the cash the assets release, measured as if all-equity.

Three layers per topic:

Layer	What it means here
U — intuition	the cash flow–production cycle, why accountants prefer historical cost, the rise in corporate cash, intangibles (Higgins Ch. 1, slides 14, 17, 26)
C — course (examinable)	the slides' definitions and every worked example: identity, NWC, book vs market, D/E, EV, liquidity ratios, the P&L lines, margins, coverage, DuPont, ROIC, P/E, the cash-flow statement, FCF and FCFE

Layer	What it means here
G — beyond the slides, flagged	linear pricing behind (1.8), the exact ROE–ROIC bridge and its volatility multiplier (5.20)–(5.22), the tax-shield term in the claims identity (6.11)–(6.13). Derived because they clarify; the course tests them through Topics 2A–2B

Reading order

Part I — the balance sheet (blocks 1.1–1.5)

1. L01-01-balance-sheet-view — slides 4–14: the three fields; the three statements; $A = L + E$ **and why every transaction preserves it**; recognition and historical cost (SVB); goods and promises, $\min(X, F) + \max(X - F, 0) = X$; $A = L + E$ **in market value derived from linear pricing**; Ross's Brasil S/A $87 = 24 + 63$; the Higgins cycle.
2. L01-02-balance-sheet-items — slides 15–21: current vs non-current; goodwill; **straight-line depreciation worked (50 m, 10 years)**; the rise in corporate cash; liabilities; NWC; **GPA line by line: NWC −842 because suppliers fund 62% of current liabilities**.
3. L01-03-book-vs-market-leverage — slides 22–31: market cap; **Starbucks: book −7.9 bn vs market 122.6 bn, and why that is not insolvency**; intangibles 17% → 84%; market-to-book; **two definitions of D/E**; net debt; **EV derived as a takeover cost (Starbucks 133.5 bn)**; current and quick ratios, and the ratio that moves the "wrong" way.

Part II — the income statement and its ratios (blocks 1.6–1.7)

4. L01-04-income-statement — slides 32–39: accrual and matching; depreciation vs expensing; gross profit → EBIT → NI; **operating leverage derived,** $1 + F/\mathbf{OI} = 2.06$ **on GM**; the statement of equity; **GM 2020 every line**, and the finance arm's interest hidden in cost of sales.
5. L01-05-ratios-dupont-roic — slides 40–51: EPS; margins; economic earnings; days; EBITDA and coverage; **DuPont derived and checked on slide 48**; Higgins's three ROE problems; **ROIC and the bridge (5.20) derived, checked on slide 50 (18% = 7.2% + 9 × 1.2%)**; "maximize leverage?" answered with (5.22), MM II and the trade-off; P/E.

Part III — cash (block 1.8)

6. L01-06-cash-flows-fcf — slides 52–62: profits ≠ cash; the three sections; **GM 2020 cash flows summed**; **slide 59's Δcash identity derived from the balance sheet**; **FCF and FCFE, GM bridge 13,290 → 18,574**; **the claims receive FCF +** $\tau \cdot$ **Int**, reconciled on Ross's numbers; what each flow values; uses of FCF; PS1 Ex. 1.

Support

7. L01-07-formulas — the content list mapped, equations (1.1)–(6.13), every number.
8. L01-08-pitfalls — 15 confusions with counterexamples, 22 original True/False items.
9. L01-09-python — the reference implementation with asserts on every worked case; runs the Lista R1 checker.

The one result to hold

$$\text{ROE} = \text{ROIC} + \frac{D}{E}\Big[\text{ROIC} - r_D(1-\tau)\Big], \qquad \text{ROIC} = \frac{\text{EBIT}(1-\tau)}{D+E}$$

ROIC is the business; the bracket is the spread; D/E scales it. Leverage raises ROE only if the business earns more than debt costs after tax, and it always multiplies ROE's swings by $1 + D/E$. Whether that higher, riskier ROE is worth anything is the question the next two lectures answer: no in a frictionless market (MM, L02), and a tax-versus-distress trade-off otherwise (L03).

Numbers and cases to have memorized

Case	Value	Meaning
Starbucks 2020	book equity **−7.9 bn**, market cap **122.6 bn**, EV **133.5 bn**	book ≠ market; negative book ≠ insolvency
GPA 2020	NWC **−842 m BRL**; payables **62%** of current liabilities	supplier finance
Slide 50	ROE **18%** vs **7.2%**; ROA **1.8%** vs **7.2%**; ROIC **7.2%** both	only ROIC is financing-neutral
Slide 48	JPMorgan **8.5% = 18.6% × 0.04 × 11.44**	leverage rescues a tiny turnover
GM 2020	**τ 21.9%**; FCF **13,290**; FCFE **18,574** (USD m)	FCF from EBIT; FCFE adds net borrowing

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Figures

Built by `figures/make_figures.py` (run: `python -I Notas/L01/figures/make_figures.py`; imports `Notas/_figstyle.py`; ends with asserts on the plotted numbers).

Figure	Shows	Lives in
L01-balance-sheets	GPA and Starbucks common-size: NWC sign, supplier finance, negative equity	L01-02-balance-sheet-items §6
L01-book-vs-market	Starbucks book vs market; EV waterfall with and without leases	L01-03-book-vs-market-leverage §3
L01-dupont	margin × turnover and ROA × leverage on log axes; slide 48 and Higgins 13e Table 2.1 firms	L01-05-ratios-dupont-roic §6
L01-roe-leverage	ROE vs D/E at ROIC 7.2% for four r_D; Companies A and B; ROA	L01-05-ratios-dupont-roic §8
L01-fcf-bridge	GM 2020: EBIT → NOPAT → FCF → FCFE	L01-06-cash-flows-fcf §6

Interactive companions

- Leverage and ROE (`companions/companion-leverage-roe.html`): five controls (EBIT, invested capital, debt share, r_D, τ) drive ROE, ROA and ROIC against D/E, a live slide-50 income statement beside its all-equity twin, and presets for Company A, Company B, the flat spread, Higgins 13e's 25% tax and a negative spread. Linked from L01-05-ratios-dupont-roic §8. Verified by `companions/check_companions.py` (page JS in node vs Python).

Scope and ceiling

- **Examinable:** every definition and worked example of slides 6–62, as stated on the slides; the derivations in notes 01–06 are the tools for applying them.
- **Named only:** SVB's HTM/AFS accounting (slide 10), Faulkender et al. (2019) (slide 17), the intangibles chart (slide 26). The mechanisms are explained, but no data beyond the slides.
- **G, not examined here:** linear pricing (1.7) (proved in L02 with state prices); (5.22); the tax term (6.11), which Topic 2B examines.
- **Out:** accounting standards beyond the slides (IFRS/GAAP detail, deferred-tax mechanics past PS1 Ex. 1, lease capitalization rules); valuation itself (Topic 2A). Scope list: the nine syllabus topics in `CLAUDE.md`.

Course-wide position

Lecture	Relation to L01
L01 (Topic 1)	the measurement layer: identity, ratios, FCF
L02 (Topic 2A)	FCF (6.6) is the valuation input; $A = D + E$ in market value (1.8) gives unlevering and MM I; ROE rising with leverage is MM II's higher cost of equity; PS2 Ex. 4 (EPS, P/E after a repurchase) uses note 05
L03 (Topic 2B)	the $\tau \cdot \text{Int}$ of (6.11) is the interest tax shield; uses of FCF (slide 62) meet dividend irrelevance; "ROE falls with more equity" is the bank-capital debate
Later topics (agency, liquidity)	what managers do with free cash; why firms hold cash (slide 17); intangibles as poor collateral

Rules pack: rules/01 covers Tirole's Introduction and Ch. 1–2 (insiders vs outsiders, stylized facts on financing). It is the syllabus framing of class 1, related to but not the content of this accounting lecture. The older notes on it are Aula/aula-01-notes.md.

Bibliography actually used in these notes

Book	Where
Slides `Aula/Topic1.pdf`	everywhere (slide numbers cited inline)

Book	Where
Higgins, Koski & Mitton, *Analysis for Financial Management*, 13e (2022), Ch. 1 pp. 3–5, 9–13, 18–28; Ch. 2 pp. 38–40, 54–57	cycle, accrual, depreciation, market vs book, cash-flow measures, imputed costs; DuPont Table 2.1, ROE problems, ROIC, P/E
Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2e (2011), Ch. 2 pp. 26–45	eqs. (2.1)–(2.5), (2.17)–(2.21); Examples 2.1–2.3; Global Corp.
Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9e (2013), Cap. 2 pp. 25, 30, 37–41; Cap. 3 pp. 60–64, 71–72	cash-flow identity [2.3] and Brasil S/A; Exemplo 3.1; Du Pont [3.24]

The slides cite Higgins 10e (2011). The repo has the 13e, whose two-firm example uses $\tau = 25\%$ instead of 40% (note 05 §8) and whose DuPont table is 2020 data, not 2013.

Related project material

- Exercise list: Lista R1 (15 exercises, three tiers), solutions PDF checked by `Listas/lista-r1-accounting-capital-structure-solutions_check.py`; roadmap roteiro-r1; study pack `Listas/study-pack-r1-accounting-capital-structure.pdf`.
- Problem set: PS1 Ex. 1(a)–(b) (`Listas/PS1.pdf`) → L01-06-cash-flows-fcf §9.
- Quiz: Topic 1 quiz.
- Audio prompt: `NotebookLM/audio/topic1-audio.md`.
- Block map: lectures-index (Topic 1); exercise ids: exercises-index; index of all units: Notas.

1.1–1.2 · The balance sheet as a split of one cash flow into promises

Claim. A firm sells two things: goods and services, and **promises** on the cash those sales generate. Creditors hold senior promises, owners hold the junior residual. Because the two claims **exhaust** the asset's cash flow, their values add up to the asset's value. That is why $A = L + E$ holds in **market** value, and not only as a bookkeeping rule. Accountants measure the same identity at historical cost, which is where book and market part ways (note 03).

Prev: — · Next: L01-02-balance-sheet-items · Map: L01-00-map

Notation.

Symbol	Meaning	Range
A, L, E	assets, liabilities, shareholders' equity (book or market, stated each time)	$A, L \geq 0$; book E can be < 0
X_t	cash flow generated by the assets at date t (state-contingent)	real
F	amount promised to creditors at a date	$F \geq 0$
$FCF_{A,t}, FCF_{L,t}, FCF_{E,t}$	cash flow of the assets, paid to creditors, paid to owners	real
PV$(\cdot)$	the market's price today of a stream of cash flows	linear (§5)

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1 · U — where corporate finance sits (slides 4–5)

Field	Object	What it hands corporate finance
Asset pricing	prices of claims to uncertain future payments	the **risk premium**, hence the discount rate for valuation (Topic 2A)
Corporate finance	governance, board structure, CEO pay; **how firms finance operations** (stocks, bonds, bank debt, convertibles); how financing affects decisions and value	—
Banking	banks and intermediaries	loans, issuance services, M&A advice to corporations

The course's question is the slide-5 sentence: how different financing choices affect the decisions of a firm and ultimately its value. This lecture builds the measurement layer that question needs.

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2 · C — three statements, two time dimensions (slide 6)

Statement	Measures	Time dimension
Balance sheet	financial condition: what the firm owns and the claims on it	**a point in time** (stock)
Income statement	financial performance: revenues and expenses	**a period** (flow)
Cash-flow statement	sources and uses of cash	**a period** (flow)

The flows link consecutive stocks: net income moves book equity (note 04, eq. (4.9)); the cash-flow statement moves the cash line (note 06, eq. (6.5)).

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3 · C — the identity and why every transaction preserves it (slides 7–9)

Assets, on the left, summarize **how the firm uses its capital**; liabilities plus equity, on the right, summarize **its sources** (slide 7). The slide-8 picture:

Assets A	Liabilities L
"Resources that generate current and future cash flows for the corporation"	"Promises made to third parties"
	Equity E — "Resources not promised away to third parties"

$$A = L + E \tag{1.1}$$

This is Berk's Balance Sheet Identity (2.1), p. 26. In the books it holds **by construction**: book equity is *defined* as the difference $A - L$ (slide 22), so (1.1) cannot fail.

Every transaction preserves it. Take differences of (1.1) between two dates:

$$\Delta A = \Delta L + \Delta E \tag{1.2}$$

The slide-9 transaction, then three more for contrast (illustrations; the exercise on this is Lista R1 Ex. 1, Higgins 1.4, p. 33):

Transaction	ΔA	ΔL	ΔE	Check of (1.2)
Buy a 1 bn plant on credit (slide 9)	$+1$	$+1$	0	$1 = 1 + 0$
Repay 1 bn of debt with cash	-1	-1	0	$-1 = -1 + 0$
Issue 1 bn of new shares for cash	$+1$	0	$+1$	$1 = 0 + 1$

Transaction	ΔA	ΔL	ΔE	Check of (1.2)
Book 0.1 bn of depreciation (no tax)	-0.1	0	-0.1 (via net income)	$-0.1 = 0 - 0.1$

Only the last row touches the income statement. The first three are pure balance-sheet moves; they change what the firm owns and owes but not what it earned. Depreciation reaches equity through net income, which is the bridge of note 04 §5.

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4 · C — what gets recognized, and at what value (slides 9–10)

Recognition. The more tangible the asset, the easier it is to value and the likelier it is to be recognized: cash, buildings, receivables. Patents, customer data, algorithms and brand reputation are measured "more conservatively", often not at all, yet some estimates put intangibles at as much as 90% of the S&P 500's total enterprise value (slide 9; the trend is slide 26, note 03 §4).

Measurement. Two bases (slide 10):

Basis	Definition	Example
Acquisition (historical) cost	the original price paid for the asset, or received for the obligation	a plant at cost less accumulated depreciation (note 02 §2)
Market value	the current market price of the asset or liability	securities that are marked to market

Silicon Valley Bank is the slide's recent example. Securities classified as **held to maturity** are carried at (amortized) historical cost; securities **available for sale** are carried at market value. When rates rise, the market value of a bond portfolio falls. The HTM part of the fall does not reach the balance sheet, so book equity overstates what the assets would fetch.

Higgins (pp. 22–23) gives the accountants' reason. Faced with "relevant but subjective current values" and "irrelevant but objective historical costs", they choose the objective numbers: "Accountants prefer to be precisely wrong than approximately right." Fair-value accounting marks some actively traded assets and liabilities to market, at the cost of more volatile reported equity.

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5 · C — two kinds of promises (slides 11–12)

A firm sells (i) goods and services, using PP&E, working capital and workers, whose sales produce the cash flows behind the **market** value of its assets. It also sells (ii) **promises**: to fund the assets it promises outside investors part of those future cash flows (slide 11).

Promise	Holder	Rank	Examples
Liabilities	non-owners (creditors)	**senior**	bonds, bank loans, payables
Equity	owners (shareholders)	**junior, residual**	stock

Owners are residual claimants: they get whatever is **not promised away** (slide 12). With one date, a cash flow $X \geq 0$ and a promise F to creditors under limited liability:

$$\text{creditors receive } \min(X, F), \qquad \text{owners receive } \max(X - F, 0) \tag{1.3}$$

The two payoffs exhaust X in every state. Case $X \geq F$:

$$\min(X, F) + \max(X - F, 0) = F + (X - F) = X$$

Case $X < F$:

$$\min(X, F) + \max(X - F, 0) = X + 0 = X$$

So in every state

$$\min(X, F) + \max(X - F, 0) = X \tag{1.4}$$

> **The choice between promises is "a key question in corporate finance" (slide 12).** Equation (1.4) already says the choice changes **who** gets the cash in each state, not **how much** cash there is. Whether it can change value is the Modigliani–Miller question of Topic 2A; the payoffs (1.3) come back there as equity being a call on the assets (L02, block 2A.5).

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6 · C — why $A = L + E$ holds in market value (slide 13)

Slide 13: "the value of an asset is the expected discounted value of its future payments", and

$$FCF_A = FCF_L + FCF_E \implies A = L + E \tag{1.5}$$

Derivation. Write the cash-flow split, date by date and state by state, as in (1.4):

$$FCF_{A,t} = FCF_{L,t} + FCF_{E,t} \quad \text{for every } t \text{ and every state} \tag{1.6}$$

Let $\mathrm{PV}(\cdot)$ be the market's price of a stream. In a market without arbitrage the price of a sum of streams is the sum of their prices (the law of one price: a portfolio of the two claims *is* the asset, so it must cost the same):

$$\mathrm{PV}(Y + Z) = \mathrm{PV}(Y) + \mathrm{PV}(Z) \tag{1.7}$$

Apply PV to both sides of (1.6):

$$\mathrm{PV}(FCF_A) = \mathrm{PV}(FCF_L + FCF_E)$$

Use (1.7) on the right-hand side:

$$\mathrm{PV}(FCF_A) = \mathrm{PV}(FCF_L) + \mathrm{PV}(FCF_E)$$

Name each price: $A^{\mathrm{mkt}} = \mathrm{PV}(FCF_A)$, $L^{\mathrm{mkt}} = \mathrm{PV}(FCF_L)$, $E^{\mathrm{mkt}} = \mathrm{PV}(FCF_E)$:

$$\boxed{A^{\mathrm{mkt}} = L^{\mathrm{mkt}} + E^{\mathrm{mkt}}} \tag{1.8}$$

Read it. The assumption doing the work is (1.7), **linear pricing**. Nothing about the split's shape matters: any F in (1.3), any seniority, any number of claims. Two caveats come later. (i) If the split changes the cash flow itself (taxes, distress costs, incentives), (1.6) no longer holds with a fixed FCF_A. Taxes are the first case: the government is a third claimant, and note 06 §7 shows the corrected identity, (6.13). (ii) Each claim is discounted at its own risk-adjusted rate, and (1.8) then forces the asset's rate to be the value-weighted average of the claims' rates. That is the unlevering formula of L02, block 2A.2.

Worked check with accounting flows (Ross, Brasil S/A, pp. 38–41). Ross's "identidade do fluxo de caixa" [2.3] (p. 37) is (1.6) measured over one year:

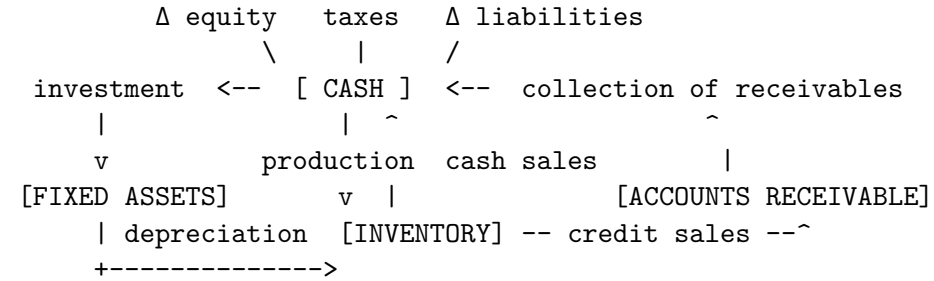
Flow (2009)	Computation	Value
Operating cash flow	EBIT + depreciation − taxes $= 694 + 65 - 212$	547
Net capital spending	ending − beginning net fixed assets + depreciation $= 1{,}709 - 1{,}644 + 65$	130
Change in NWC	$1{,}014 - 684$	330
Cash flow from assets	$547 - 130 - 330$	**87**
Cash flow to creditors	interest − net new borrowing $= 70 - 46$	24
Cash flow to shareholders	dividends − new equity $= 103 - 40$	63
Sum of the claims	$24 + 63$	**87**

The identity checks: $87 = 24 + 63$**.** Ross's operating cash flow subtracts **actual** taxes, so his cash flow from assets already contains the tax saved by deducting interest. The slides' unlevered FCF (slide 60) does not, and the two differ by exactly $\tau\times$interest. Note 06 §7 reconciles them on these same numbers.

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7 · U — the cash flow–production cycle (slide 14)

Slide 14 reproduces Higgins's Figure 1.1 (13e, p. 4):



Cash becomes inventory through production. Inventory becomes cash directly, through cash sales, or later, through credit sales and the collection of receivables. That loop is the **working-capital cycle**. Investment turns cash into fixed assets, and depreciation passes "a small portion of the value of fixed assets" into each unit of inventory (Higgins pp. 4–5). Higgins draws two principles from the figure (p. 5):

1. **Financial statements are a window on operations.** Credit terms, production technique and inventory control set the firm's financial profile. Tighter collection shows up as lower receivables.
2. **Profits do not equal cash flow.** A firm that lets customers pay ever later, or keeps producing more than it sells, can be profitable to the accountant and still run out of cash. This is the subject of note 06.

Links

- Next: the line items of the identity and NWC → L01-02-balance-sheet-items; book vs market → L01-03-book-vs-market-leverage; the cash-flow side of (1.6) → L01-06-cash-flows-fcf.
- Forward: value additivity and MM I in L02 (blocks 2A.5–2A.8); the tax claimant and $V_L = V_U + \tau D$ in L03.
- Block map: lectures-index Topic 1, blocks 1.1–1.2.
- Rules pack: rules/01 (Tirole Intro, Ch. 1–2: insiders vs outsiders, value additivity at Tirole p. 78; related framing, not this lecture's accounting).
- Practice: Lista R1 Ex. 1 (Higgins 1.4, p. 33); quiz Topic 1 quiz tag `bs`.
- Books: Berk 2e Ch. 2 §2.2 (pp. 26–29); Higgins 13e Ch. 1 (pp. 3–5, 22–23); Ross *Fundamentos* Cap. 2 §2.1 (p. 25), §2.4 (pp. 37–41).

1.3 · Balance-sheet items: current vs non-current, depreciation, net working capital

> **Claim.** The balance sheet sorts assets and liabilities by **horizon**: within one year, or beyond. Fixed assets enter at cost and are written down by depreciation, an accounting allocation that **moves no cash**. Current assets minus current liabilities is **net working capital**, the short-term cushion. The GPA example shows NWC can be negative because **suppliers finance the firm**.

Prev: L01-01-balance-sheet-view · Next: L01-03-book-vs-market-leverage · Map: L01-00-map

Notation.

Symbol	Meaning	Range
CA, CL	current assets, current liabilities	≥ 0
NWC	net working capital, $CA - CL$	real
K_0	acquisition cost of a fixed asset	> 0
S_v	salvage value at the end of its life	$0 \leq S_v \leq K_0$
n	productive life, years	$n \geq 1$
Dep_k, BV_k	depreciation in year k, book value at the end of year k	≥ 0

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1 · C — current assets (slide 15)

Current assets can be converted into cash **within one year** (slide 15; Berk §2.2, p. 27):

Item	What it is
Cash and marketable securities	short-term, liquid, low-risk investments: money-market funds, government debt, bank deposits
Accounts receivable	owed by customers who bought on credit
Inventories	raw materials, work in progress, finished goods

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2 · C — non-current assets, goodwill and depreciation (slide 16)

Non-current ("long-term") assets: fixed assets such as property, plant and equipment, and **goodwill**. Slide 16 defines goodwill as the difference between the price of an acquisition and the *fair value* of what was acquired. Fair value is "the accountant's estimate of the potential value of an asset, not necessarily equal to book value":

$$\text{Goodwill} = \text{price paid for the target} - \text{fair value of its identifiable net assets} \tag{2.1}$$

Depreciation. Fixed assets wear out; the accountant writes down their book value by an amount called depreciation. Two facts carry through the whole lecture (slide 16):

Depreciation is not a cash expense. The cash left the firm when the asset was bought. **Book value of an asset = acquisition cost − accumulated depreciation.**

With straight-line depreciation (one of the methods of slide 35; note 04 §3):

$$\text{Dep}_k = \frac{K_0 - S_v}{n} \quad (k = 1, \dots, n) \tag{2.2}$$

$$BV_k = K_0 - \sum_{j=1}^{k} \text{Dep}_j \tag{2.3}$$

Worked example (Higgins 13e, p. 13). A facility costs $K_0 = 50$ million, has a 10-year life, and, we add for the arithmetic, no salvage value. Substituting $K_0 = 50$, $S_v = 0$, $n = 10$ into (2.2):

$$\text{Dep}_k = \frac{50 - 0}{10} = 5 \text{ million per year}$$

Substituting into (2.3) at the end of year 3:

$$BV_3 = 50 - (5 + 5 + 5) = 35 \text{ million}$$

End of year k	0	1	3	5	10
BV_k (millions)	50	45	35	25	0
Cash paid in year k	50	0	0	0	0

The book value falls by 5 a year while the cash row is zero after year 0. The 5 per year is an allocation of a past outlay to the years it serves (Higgins: "the allocation of past expenditures to future time periods to match revenues and expenses", p. 13). Note 04 §3 shows the income-statement side. Note 06 §2 shows why the cash-flow statement adds the 5 back.

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3 · U — the rise in corporate cash (slide 17)

Slide 17 reproduces Figure 1 of Faulkender, Hankins and Petersen (2019), built from the Federal Reserve Flow of Funds, Table L.102 (nonfinancial business): checking and savings accounts plus debt securities held, 1994–2017. Read off the chart, aggregate corporate cash rises from under 1 trillion dollars to roughly 4 trillion, and **faster than GDP**: the cash/GDP index goes from 100 (1994) to about 160. The paper's title states the two candidate explanations: **precautionary savings** or **foreign taxes**. Why firms hold liquidity at all is the liquidity topic of the syllabus (rules/05).

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4 · C — liabilities (slides 18–19)

Current liabilities, due within one year (slide 18):

Item	What it is
Debt due next year	from short-term **and** long-term instruments (current portion of long-term debt)
Accounts payable	owed to suppliers for credit purchases
Accruals	salaries, taxes owed but not yet paid
Deferred (unearned) revenue	cash received for products not yet delivered

Non-current liabilities extend beyond one year: debt payments not due within the next accounting year (slide 19).

Berk (p. 28) adds a distinction that matters later. Notes payable and short-term debt come from **financing** decisions; accounts payable and accruals come from **operating** decisions. Free cash flow (note 06, eq. (6.6)) puts the operating liabilities in working capital and the financing ones in debt.

---

5 · C — net working capital (slide 20)

$$NWC = CA - CL \tag{2.4}$$

This is Berk's (2.2), p. 28. It measures short-term financial health: firms with low NWC "might not have enough resources to meet short-term obligations" (slide 20).

Berk's Global Corporation, 2010 (p. 28). Substituting $CA = 57.0$, $CL = 34.7$ into (2.4):

$$NWC = 57.0 - 34.7 = 22.3 \text{ million}$$

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6 · C — worked example: GPA, 31/12/2020 (slide 21)

BRL millions, as on the slide:

Assets		Liabilities & shareholders' equity	
Current assets	**17,641**	**Current liabilities**	**18,483**
Cash and equivalents	8,711	Trade payable, net	11,424
Trade receivable, net	686	Borrowings and financing	2,309
Inventories	6,536	Lease liabilities	947
Other	1,708	Financing of property and equipment	100
		Other	3,703
Non-current assets	**35,654**	**Non-current liabilities**	**18,005**
Property and equipment, net	19,888	Borrowings and financing	6,842
Intangible	6,164	Lease liabilities	7,427
Other	9,602	Other	3,736

Assets		Liabilities & shareholders' equity	
		Total liabilities	**36,488**
		Shareholders' equity	**16,807**
Total assets	**53,295**	**Total liabilities & equity**	**53,295**

Every subtotal checks.

$$CA = 8{,}711 + 686 + 6{,}536 + 1{,}708 = 17{,}641$$

$$NCA = 19{,}888 + 6{,}164 + 9{,}602 = 35{,}654$$

$$A = 17{,}641 + 35{,}654 = 53{,}295$$

$$CL = 11{,}424 + 2{,}309 + 947 + 100 + 3{,}703 = 18{,}483$$

$$NCL = 6{,}842 + 7{,}427 + 3{,}736 = 18{,}005$$

$$L = 18{,}483 + 18{,}005 = 36{,}488$$

$$L + E = 36{,}488 + 16{,}807 = 53{,}295 = A \quad \checkmark \ (1.1)$$

Net working capital, substituting into (2.4):

$$NWC = 17{,}641 - 18{,}483 = -842 \text{ million BRL}$$

Where the short-term money comes from. Trade payables relative to current liabilities and to total assets:

$$\frac{11{,}424}{18{,}483} = 0.6181, \qquad \frac{11{,}424}{53{,}295} = 0.2144$$

Suppliers fund 62% of GPA's current liabilities and 21% of its whole balance sheet, against 2,309 of short-term borrowings. This is the slide's remark ("note the importance of suppliers in the provision of short-term financing"). A grocer sells for cash quickly and pays suppliers later, so **negative NWC is a feature of the business model, not a sign of distress**. Whether trade credit is cheap finance or a symptom of constraints is the subject of Lista R5b.

Slides 21 and 24 as percentages of total assets; equation (1.1) is the dotted 100% line; Starbucks' liabilities run past 100% and its equity hangs below zero.

Both firms satisfy (1.1). GPA's two columns meet at 100% with equity on top. At Starbucks the liabilities alone reach 126% of the assets, so book equity must be −26%: the identity forces a negative residual. Payables are 21% of GPA's balance sheet and 3.5% of Starbucks'. Note 03 takes up Starbucks.

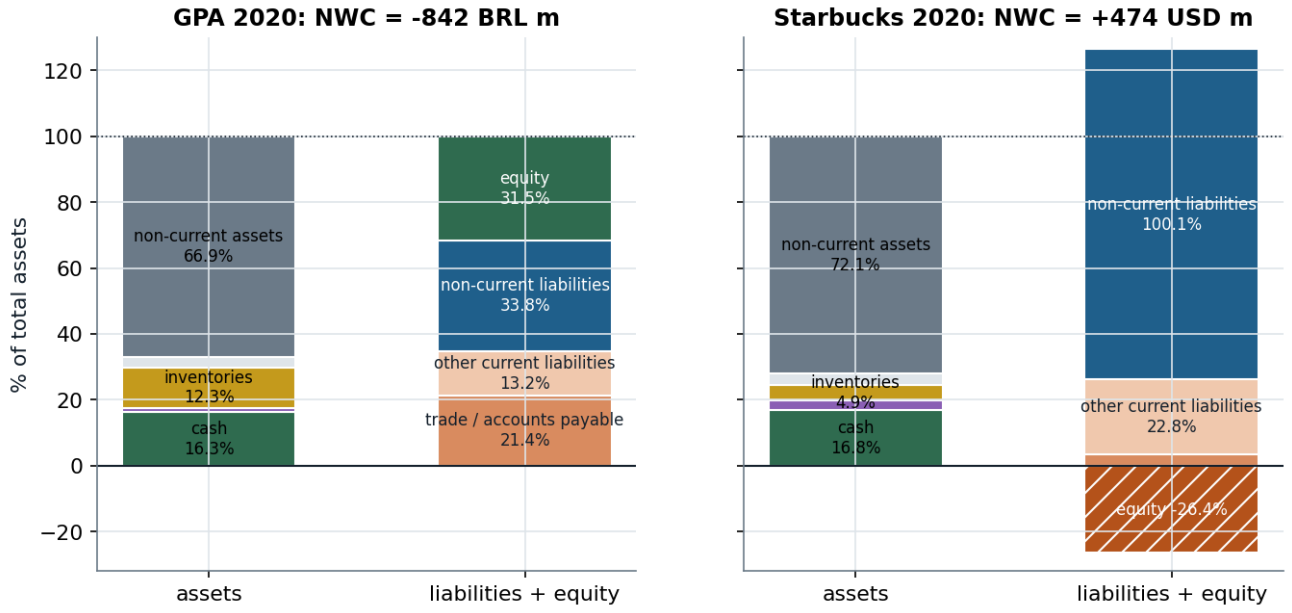


Figure 1: Common-size balance sheets of GPA and Starbucks, 2020: assets column (cash, receivables, inventories, other current, non-current) beside the liabilities-plus-equity column (trade payables, other current liabilities, non-current liabilities, equity), each as a percent of total assets; GPA's equity is 31.5% while Starbucks' equity is a negative 26.4% below the zero line

Links

- Prev / next: L01-01-balance-sheet-view · L01-03-book-vs-market-leverage (current and quick ratios on these same numbers, §8).
- Depreciation in the income statement: L01-04-income-statement §3; in the cash-flow statement: L01-06-cash-flows-fcf §2.
- NWC in free cash flow: L01-06-cash-flows-fcf eq. (6.6); NWC recouped at a project's end: L02 block 2A.3.
- Rules: rules/05 (liquidity, why firms hold cash).
- Practice: Lista R1 Ex. 1 (Higgins 1.4, p. 33), Ex. 14 (Higgins 2.11, p. 74, balance-sheet reconstruction); quiz Topic 1 quiz tag `bs`.
- Books: Berk 2e §2.2 (pp. 26–28); Higgins 13e Ch. 1 "The Balance Sheet" (pp. 9–10), "Depreciation" (p. 13).

1.4–1.5 · Book vs market equity, leverage ratios, enterprise value, liquidity ratios

> **Claim.** Book equity is a residual of historical costs. Market capitalization is a price on **future** cash flows, including those of assets the balance sheet never records. The two can differ in sign: Starbucks had book equity of **−7.9 bn USD** and a market cap of **122.6 bn USD**, and it was not insolvent. Leverage and value ratios therefore use **market** equity. **Enterprise value** prices the business alone: equity plus debt minus cash.

Prev: L01-02-balance-sheet-items · Next: L01-04-income-statement · Map: L01-00-map

Notation.

Symbol	Meaning	Range
E^{book}, E^{mkt}	book equity ($A - L$); market capitalization	book real, market ≥ 0
P, N	share price; common shares outstanding	> 0
D	interest-bearing debt (short- and long-term, incl. current maturities), book value	≥ 0
L	total liabilities	$\geq D$
C	cash and marketable securities	≥ 0
EV	enterprise value	real
CA, CL, Inv	current assets, current liabilities, inventories	≥ 0

---

1 · C — book equity and market capitalization (slide 22)

$$E^{\text{book}} = A - L \tag{3.1}$$

$$E^{\text{mkt}} = P \times N \tag{3.2}$$

Book equity is "an accounting measure of the net worth of the firm". Market capitalization is what investors are willing to pay for the equity. Slide 22 sums up the difference: **market values are forward-looking** (they reflect the expected discounted value of future payoffs to shareholders), **book values are backward-looking**.

Berk, Example 2.1 (p. 29). Global has $N = 3.6$ million shares at $P = 10.00$. Substituting into (3.2):

$$E^{\text{mkt}} = 10.00 \times 3.6 = 36 \text{ million}$$

against $E^{\text{book}} = 22.2$ million. **Higgins, Polaris (p. 25):** $E^{\text{mkt}} = 95.28 \times 61.9 = 5{,}898$ million (rounded as printed), against a book value of 1,434 million. The ratio is $5{,}898/1{,}434 = 4.11$.

---

2 · C — why book and market differ (slide 23)

Driver	Mechanism	Source
Historical cost	assets are carried at cost less depreciation; the cost does not update for changes in the income the asset will generate	slide 23; Higgins pp. 22–23
Little marking to market	only a few assets and liabilities trade actively	slide 23
Off-balance-sheet assets	quality of employees and managers, future R&D, reputation, loyal customers	slide 23; Berk p. 29
Off-balance-sheet liabilities	pending lawsuits (product-liability claims)	slide 23
What equity buys	"equity investors buy shares for the future income they hope to receive, not for the value of the firm's assets"	Higgins p. 23

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3 · C — worked example: Starbucks, 27/12/2020 (slides 24–25)

USD millions, as on slide 24:

Assets		Liabilities & shareholders' equity	
Current assets	**8,358**	**Current liabilities**	**7,884**
Cash and equivalents	5,028	Accounts payable	1,051
Accounts receivable, net	888	Accrued payroll and benefits	685
Inventories	1,472	Current portion of operating lease liability	1,268
Other	970	Short-term debt	493
		Current portion of long-term debt	750
		Stored value card liability and current portion of deferred revenue	1,871
		Other	1,767
Non-current assets	**21,611**	**Non-current liabilities**	**29,989**
Property and equipment, net	6,118	Long-term debt	14,674
Operating lease, right-of-use asset	8,199	Operating lease liability	7,755
Goodwill	3,707	Other	7,561

Assets		Liabilities & shareholders' equity	
Other	3,587	**Total liabilities**	**37,872**
		Shareholders' equity (deficit)	**(7,904)**
Total assets	**29,968**	**Total liabilities & equity**	**29,968**

The identity, then the residual. Substituting into (3.1):

$$E^{\text{book}} = 29{,}968 - 37{,}872 = -7{,}904 \text{ million}$$

(The line items are rounded: they sum to 7,885 and 29,990 against the printed subtotals of 7,884 and 29,989, a one-unit rounding difference.)

Is Starbucks insolvent? Slide 25: **no.** Book equity is −7.9 bn and the market capitalization on 28 December 2020 was 122.6 bn:

$$E^{\text{mkt}} - E^{\text{book}} = 122.6 - (-7.9) = 130.5 \text{ bn USD}$$

> **How does book equity go negative?** The statement of equity (note 04, eq. (4.9)) moves book equity by net income minus dividends, plus or minus issuance and **repurchases**. When cumulative payouts exceed paid-in capital plus retained earnings, the residual turns negative, while the brand, the customers and the store network that produce the cash stay off the balance sheet. Creditors lend against the **market** value of the assets: Berk (p. 29) notes that successful firms "are often able to borrow in excess of the book value of their assets". Amazon in June 2005 had book equity of −64 million and a market value of equity over 15 billion (p. 29).

Slides 24–25 (book items, market cap 122.6 bn) and eq. (3.6). Debt = short-term debt + current portion of LTD + long-term debt = 15.9 bn; the lease column is a convention choice, not on the slide.

The left panel shows the sign flip: the book claim of the owners is negative and the market's claim is four times total book assets. The right panel builds enterprise value from the same balance sheet (§7). Whether operating-lease liabilities count as debt changes EV by 9 bn, about 7%.

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4 · U — intangible assets (slide 26)

Slide 26 (Visual Capitalist; "Tangible Assets vs. Intangible Assets for S&P 500 Companies, 1975–2018") stacks tangible and intangible value. The intangible share, computed from the chart's own labels:

Year	Total	Intangible	Intangible share
1975	715 bn	122 bn	122/715 = 17.1%
1985	1.5 tn	482 bn	0.482/1.5 = 32.1%
1995	4.59 tn	3.12 tn	3.12/4.59 = 68.0%
2005	11.6 tn	9.28 tn	9.28/11.6 = 80.0%
2018	25.03 tn	21.03 tn	21.03/25.03 = 84.0%

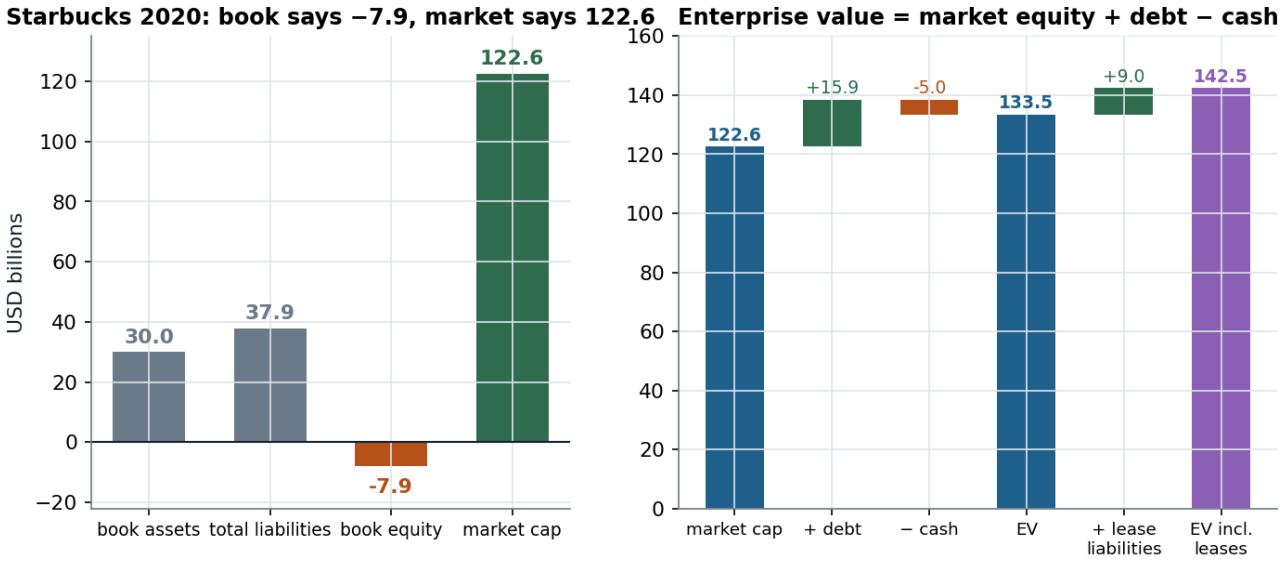


Figure 2: Left: Starbucks 2020 book assets 30.0, total liabilities 37.9, book equity −7.9 and market capitalization 122.6, in USD billions. Right: a waterfall from market cap 122.6 plus interest-bearing debt 15.9 minus cash 5.0 to an enterprise value of 133.5, then plus lease liabilities 9.0 to 142.5

> **Tangibles are "easy to value, thick and efficient secondary markets, insurable". Intangibles are "difficult to value, thin and inefficient secondary markets, difficult to insure"** (slide 26 legend). The same properties that keep intangibles off the balance sheet make them poor collateral. This is the reason "redeployable assets" support high leverage in Tirole's stylized facts (rules/01).

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5 · C — the market-to-book ratio (slide 27)

$$\text{Market-to-book} = \frac{E^{\text{mkt}}}{E^{\text{book}}} \tag{3.3}$$

This is Berk's (2.3), p. 30. **Growth stocks** have a high market-to-book ratio; **value stocks** a low one (slide 27). Berk's 2010 figures (p. 30): AmeriServ Financial 0.52, Citigroup 0.82, financial services about 1, large U.S. firms close to 3, technology about 6, Amazon over 11.

Starbucks. Substituting into (3.3): $122{,}600/(-7{,}904) = -15.5$. **A negative ratio carries no information about growth.** The ratio is meaningful only when $E^{\text{book}} > 0$.

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6 · C — the debt-equity ratio, book and market (slides 28–29)

The slide's definition (slide 28):

$$\left.\frac{D}{E}\right|_{\text{slide}} = \frac{L}{E} \tag{3.4}$$

Berk's (2.4) (p. 30) uses **interest-bearing debt** in the numerator ("short- and long-term debt, including current maturities"). Either can be computed with book or market equity. **Always state which definition you use.**

Firm	Numerator	Book E	Market E	Book ratio	Market ratio
Global 2010 (Berk p. 31)	$D = 5.5 + 113.2 = 118.7$	22.2	36	$118.7/22.2 = 5.3$	$118.7/36 = 3.3$
Starbucks 2020	$L = 37{,}872$	$-7{,}904$	122,600	-4.79 (meaningless)	$37{,}872/122{,}600 = 0.309$
Starbucks 2020	$D = 493 + 750 + 14{,}674 = 15{,}917$	$-7{,}904$	122,600	-2.01 (meaningless)	$15{,}917/122{,}600 = 0.130$
GPA 2020	$L = 36{,}488$	16,807	—	$36{,}488/16{,}807 = 2.171$	—

> **The market ratio "is more informative about the firm's leverage and capacity to meet future debt payments"** (slide 28). Even firms with negative book equity may carry a lot of debt because they are expected to be profitable, which makes the market ratio "particularly useful for assessing the leverage of high-growth and start-up firms". Berk (p. 31, fn. 3) adds that the book value of debt is used as a proxy for its market value because the two are usually close (slide 29 says the same).

Net debt treats cash as negative debt, "used on short notice to repay debts" (slide 29):

$$\text{Net debt} = D - C \tag{3.5}$$

The slide writes "liabilities less cash"; with interest-bearing debt, Starbucks: $15{,}917 - 5{,}028 = 10{,}889$ million. GPA, using borrowings only: $(2{,}309 + 6{,}842) - 8{,}711 = 9{,}151 - 8{,}711 = 440$ million BRL. Net debt is the better measure for firms with large cash balances.

---

7 · C — enterprise value (slide 30)

$$EV = E^{\text{mkt}} + D - C \tag{3.6}$$

This is Berk's (2.5), p. 31. **Derivation as a takeover cost** (slide 30's question: "how much cash does an investor need to be the single owner of all the assets?"). The buyer pays three amounts:

$$\text{buy all the equity} : +E^{\text{mkt}}$$

$$\text{pay off the debt} : +D$$

$$\text{receive the firm's cash} : -C$$

Summing the three gives (3.6): what is bought is the business assets, net of the cash that comes with them.

Worked cases.

Berk, Global (p. 31). Substituting $E^{\text{mkt}} = 36$, $D = 118.7$, $C = 23.2$ into (3.6):

$$EV = 36 + 118.7 - 23.2 = 131.5 \text{ million}$$

Berk, Example 2.2, Heinz, April 2010 (p. 32). The market cap, then book equity from the market-to-book ratio (3.3), then debt from the book D/E:

$$E^{\text{mkt}} = 46.15 \times 316.2 \text{ m} = 14.59 \text{ bn}$$
$$E^{\text{book}} = 14.59/7.99 = 1.83 \text{ bn}$$
$$D = 1.83 \times 2.64 = 4.83 \text{ bn}$$
$$EV = 14.59 + 4.83 - 0.5623 = 18.858 \text{ bn}$$

Starbucks, from slides 24–25. Substituting $E^{\text{mkt}} = 122{,}600$, $D = 15{,}917$, $C = 5{,}028$:

$$EV = 122{,}600 + 15{,}917 - 5{,}028 = 133{,}489 \text{ million USD}$$

Counting the operating-lease liabilities ($1{,}268 + 7{,}755 = 9{,}023$) as debt gives $133{,}489 + 9{,}023 = 142{,}512$. Both are in the figure of §3.

> **EV values the business; market cap values one claim on it.** EV is what FCF forecasts value (slide 61, note 06 §8). Because it nets out financing, EV multiples (EV/EBITDA, Berk pp. 39–40) are comparable across firms with different leverage, and P/E is not (note 05 §10).

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8 · C — current and quick ratios (slide 31)

$$\text{Current ratio} = \frac{CA}{CL} \tag{3.7}$$

$$\text{Quick ratio (acid test)} = \frac{CA - \text{Inv}}{CL} \tag{3.8}$$

Both measure the ability to meet short-term obligations; "some debt covenants require the current ratio to be above a certain value" (slide 31).

Firm	Current ratio (3.7)	Quick ratio (3.8)
GPA 2020	$17{,}641/18{,}483 = 0.954$	$(17{,}641 - 6{,}536)/18{,}483 = 11{,}105/18{,}483 = 0.601$
Starbucks 2020	$8{,}358/7{,}884 = 1.060$	$(8{,}358 - 1{,}472)/7{,}884 = 6{,}886/7{,}884 = 0.873$

Ross, Exemplo 3.1 (p. 62): paying current liabilities with cash pushes the ratio away from 1. Start from $CA = 4$, $CL = 2$ and use 1 of cash to pay 1 of current liabilities:

$$\frac{4-1}{2-1} = 3 > 2$$

Start instead from $CA = 2$, $CL = 4$:

$$\frac{2-1}{4-1} = \frac{1}{3} < \frac{1}{2}$$

A ratio is not a level of liquidity. The same transaction raises a healthy firm's current ratio and lowers a stressed firm's. Buying inventory for cash leaves (3.7) unchanged but lowers (3.8) (Ross p. 62).

Links

- Prev / next: L01-02-balance-sheet-items · L01-04-income-statement.
- Market equity in returns ratios: L01-05-ratios-dupont-roic §7 (the value problem), §10 (P/E).
- EV and FCF: L01-06-cash-flows-fcf §8. Market-value D/E as the input of unlevering betas and MM II: L02 blocks 2A.2, 2A.9.
- Rules: rules/01 (Tirole Ch. 2 stylized facts on leverage); rules/02 (valuation).
- Practice: Lista R1 Ex. 4 (Berk 2.8, p. 53: market cap, market-to-book, D/E, EV), Ex. 5 (Higgins 2.3, p. 71: ratio identities), Ex. 6 (Berk 2.21, p. 55: book vs market leverage); quiz Topic 1 quiz tag `bs`.
- Books: Berk 2e §2.2–2.3 (pp. 29–33); Higgins 13e Ch. 1 pp. 22–25; Ross *Fundamentos* Cap. 3 §3.3 (pp. 60–64).

1.6 · The income statement: accrual, matching, depreciation, from sales to net income

> **Claim.** The income statement measures **profit over a period**, and profit is an **allocation**, not a count of cash. Revenue is booked when the sale is made, costs when goods and services are bought, and a fixed asset's cost is spread over its life. The lines from sales to net income show **who is paid in what order**: suppliers and workers, then lenders (interest), then the state (taxes), then owners. A cost structure heavy in fixed costs makes operating income move **more than proportionally** with sales.

Prev: L01-03-book-vs-market-leverage · Next: L01-05-ratios-dupont-roic · Map: L01-00-map

Notation.

Symbol	Meaning	Range
S	sales (revenue) of the period	≥ 0
COGS	cost of goods sold (cost of sales)	≥ 0
OpEx	other operating expenses (SG&A, R&D, depreciation)	≥ 0
NOI	nonoperating income	real
Int, T	interest expense, income taxes	≥ 0
v, F	variable-cost share of sales, fixed costs (§4)	$0 \leq v < 1$, $F \geq 0$
ΔAR	increase in accounts receivable over the period	real
Div	dividends paid	≥ 0

---

1 · C — what the income statement measures (slide 32)

The income statement (P&L) "attempts to measure the profits the firm generates during a period of time": it lists revenues and expenses. **Revenues increase book equity, expenses reduce it** (slide 32). Through net income it is the flow that links two balance sheets (§5).

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2 · C — the accrual and matching principles (slides 32–33)

Accrual principle: revenue is recognized at the time of the sale, not when the customer pays; expenses when goods and services are bought, not when suppliers are paid. **Matching principle:** revenues and expenses are recorded in the period in which the transaction occurs (slide 32).

The slide-33 table:

Event	Cash flow?	Revenue / expense?
A customer buys and pays in cash	Yes	Yes
A customer pays the bill for an earlier purchase	Yes	No
A customer buys and agrees to pay later	No	Yes

The gap between the two columns is the change in receivables:

$$\text{cash collected from customers} = S - \Delta AR \tag{4.1}$$

Higgins, Polaris 2020 (pp. 12–13). Revenue 7,108 million; receivables rose by 67 million. Substituting into (4.1):

$$\text{cash collected} = 7{,}108 - 67 = 7{,}041 \text{ million}$$

"The other 67 million still awaits collection."

---

3 · C — depreciation: spreading capex over the asset's life (slides 34–35)

Charging a fixed asset's full cost in the year of purchase would make profit "too low in the year of the investment, and too high in the other productive years" (slide 34). Spreading the cost as depreciation **matches** the asset's cost to the revenue it helps produce.

Higgins's facility (p. 13), with eq. (2.2) of note 02: cost 50 million, 10-year life, straight line, no salvage value, so 5 million a year. The two treatments of the same outlay, years 1 to 3 (the purchase is booked in year 1 under expensing):

Year	1	2	3	…	10	Sum over 10 years
Expense if charged at once	50	0	0	…	0	50
Depreciation (straight line)	5	5	5	…	5	50
Cash outflow	50	0	0	…	0	50

> **Both treatments expense 50 in total and neither changes the cash row.** Depreciation changes **when** profit falls, not **whether** cash leaves. "Depreciation is not a cash expense; the cash outlay only occurs at the time of the investment" (slide 34). Its one cash consequence runs through taxes, since depreciation is deductible (note 06 §6, note 08 pitfall 3).

How much to depreciate requires three estimates: productive life, salvage value, and the method, straight line or accelerated (slide 35; Higgins p. 13). Also by the matching principle, **an increase in inventory is not an expense**: goods are expensed (as COGS) when sold (slide 35).

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4 · C — the lines of the income statement and operating leverage (slides 36–37)

From sales to net income, one subtraction per line:

$$\text{Gross profit} = S - \text{COGS} \tag{4.2}$$

$$\text{Operating income} = \text{Gross profit} - \text{OpEx} \tag{4.3}$$

$$\text{EBIT} = \text{Operating income} + \text{NOI} \tag{4.4}$$

$$\text{Net income} = \text{EBIT} - \text{Int} - T \tag{4.5}$$

EBIT is income "before it is split among debtholders, equityholders, and the tax authorities" (slide 37). Net income is the owners' residual. Apart from issuance or payout, it is what makes book equity grow (§5).

Operating leverage. Slide 36: most COGS is **variable** (moves with sales), most other operating expenses are **fixed**, and "companies with a high proportion of fixed costs are more vulnerable to sale declines". To see why, write COGS $= vS$ and OpEx $= F$ in (4.2)–(4.3):

$$\text{Operating income} = S - vS - F = (1 - v)S - F \tag{4.6}$$

A change ΔS in sales, with v and F fixed, changes (4.6) by

$$\Delta\,\text{Operating income} = (1 - v)\,\Delta S \tag{4.7}$$

Divide the percentage change in operating income by the percentage change in sales:

$$\frac{\Delta\text{OI}/\text{OI}}{\Delta S/S} = \frac{(1-v)\Delta S/\text{OI}}{\Delta S/S} = \frac{(1-v)S}{\text{OI}}$$

Substitute $(1 - v)S = \text{OI} + F$ from (4.6):

$$\frac{\Delta\text{OI}/\text{OI}}{\Delta S/S} = \frac{\text{OI} + F}{\text{OI}} = 1 + \frac{F}{\text{OI}} \tag{4.8}$$

The elasticity of operating income to sales exceeds 1 by the ratio of fixed costs to operating income. With $F = 0$ the elasticity is 1. A firm whose fixed costs are large relative to its operating income sees its profit swing by a multiple of its sales.

On GM's numbers (slide 39, §6), with the slide's own heuristic (all of cost of sales variable, all of SG&A fixed): $F = 7{,}038$, $\text{OI} = 6{,}634$. Substituting into (4.8):

$$1 + \frac{7{,}038}{6{,}634} = 1 + 1.0609 = 2.061$$

A 10% fall in sales cuts gross profit from 13,672 to $0.9 \times 13{,}672 = 12{,}304.8$. Operating income falls to

$$12{,}304.8 - 7{,}038 = 5{,}266.8$$

which is a change of

$$\frac{5{,}266.8 - 6{,}634}{6{,}634} = -20.61\% = 2.061 \times (-10\%)$$

---

5 · C — the statement of equity (slide 38)

$$E^{\text{book}} = \text{paid-in capital} + \text{retained earnings}$$

$$E^{\text{book}}_{\text{end}} = E^{\text{book}}_{\text{begin}} + \text{NI} - \text{Div} \pm \text{other changes} \tag{4.9}$$

"Other changes" are equity issuance, treasury-share transactions (repurchases) and stock-based compensation (slide 38). With no other changes, retained earnings grow by NI − Div, Berk's (2.20) (p. 44). **Berk's Global, 2010 (p. 44):** NI 2 million, dividends 1 million, so retained earnings rise by $2 - 1 = 1$ million, a payout ratio of $1/2 = 50\%$ (Berk (2.21)).

> **(4.9) explains Starbucks' negative book equity (note 03 §3):** large repurchases enter "other changes" with a minus sign, and nothing in (4.9) requires the residual to stay positive.

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6 · C — worked example: GM, income statement 2020 (slide 39)

USD millions, as on slide 39, with each subtotal recomputed:

Line	Value	Check
Net sales automotive	108,673	
Net sales Financial	13,812	
Net sales	**122,485**	108,673 + 13,812 = 122,485
Automotive and other cost of sales	97,539	
GM financial interest, operating and other expenses	11,274	
Cost of sales	**108,813**	97,539 + 11,274 = 108,813
Gross profit (4.2)	**13,672**	122,485 − 108,813 = 13,672
Automotive and other SG&A	7,038	
Operating income (4.3)	**6,634**	13,672 − 7,038 = 6,634
Nonoperating income	2,559	
EBIT (4.4)	**9,193**	6,634 + 2,559 = 9,193
Interest expense	1,098	
Pre-tax income	**8,095**	9,193 − 1,098 = 8,095
Taxes	1,774	
Net income (4.5)	**6,321**	8,095 − 1,774 = 6,321

Effective tax rate (used for GM's free cash flow in note 06 §6):

$$\tau = \frac{1{,}774}{8{,}095} = 0.21915$$

> **Read the cost-of-sales line carefully.** GM Financial's **interest** sits inside cost of sales (11,274, "GM financial interest, operating and other expenses"). For a captive finance arm, borrowing is part of operations, so the "interest expense" line (1,098) is only the automotive side's. Ratios that separate operating from financing items (EBIT margin, coverage, ROIC, FCF) are blurred for firms with a finance subsidiary.

Links

- Prev / next: L01-03-book-vs-market-leverage · L01-05-ratios-dupont-roic (margins, EBITDA and coverage on these GM numbers, §2 and §5).
- Depreciation and accruals in cash flow: L01-06-cash-flows-fcf §2; book value of an asset: L01-02-balance-sheet-items §2.
- Forward: taxable income and the interest tax shield in L03 (block 2B.1).
- Practice: Lista R1 Ex. 2 (Berk 2.12, p. 54: margins), Ex. 3 (Berk 2.31, p. 57: depreciation, earnings vs cash), Ex. 15 (Ross 2.25–2.26, pp. 50–51: two years of statements); quiz Topic 1 quiz tag `is`.
- Books: Higgins 13e Ch. 1 "The Income Statement", "Accrual Accounting", "Depreciation" (pp. 11–13); Berk 2e §2.4 (pp. 33–34); Ross *Fundamentos* Cap. 2 §2.2 (p. 30).

1.7 · Income-statement analysis: margins, coverage, DuPont, ROE vs ROIC

Claim. Every ratio here answers one narrow question, and most mix **operations** with **financing**. The DuPont identity splits ROE into margin × turnover × leverage, so ROE can rise purely because equity shrinks. ROIC strips the financing out. The exact bridge between the two is $\text{ROE} = \text{ROIC} + \frac{D}{E}[\text{ROIC} - r_D(1-\tau)]$: leverage adds the **spread** times D/E, and it multiplies ROE's sensitivity to ROIC by $1 + D/E$. That is why the slide's question "should managers maximize financial leverage?" has no accounting answer.

Prev: L01-04-income-statement · Next: L01-06-cash-flows-fcf · Map: L01-00-map

Notation.

Symbol	Meaning	Range
NI, S	net income, sales	real, ≥ 0
N	shares outstanding	> 0
A, E, L	total assets, book equity, total liabilities	$A > 0$, $E > 0$ here
D	interest-bearing debt	≥ 0
r_D	interest rate on debt (interest $= r_D D$)	≥ 0
τ	tax rate on pre-tax income (full loss offset assumed in §8)	$[0, 1)$
r_E	required return on equity (hurdle rate)	> 0
ROIC	return on invested capital, (5.16)	real

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1 · C — earnings per share, basic and diluted (slide 40)

$$\text{EPS} = \frac{\text{NI}}{N} \tag{5.1}$$

$$\text{EPS}^{\text{diluted}} = \frac{\text{NI}}{N + N^{\text{options}} + N^{\text{convertibles}}} \tag{5.2}$$

Stock options for employees and convertible bonds can raise the share count; that growth is **dilution**. Diluted EPS assumes all options exercised and all convertibles converted (slide 40). **Berk, Global 2010 (p. 34):** EPS $= 2.0/3.6 = 0.56$ per share.

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2 · C — margins (slides 41, 43)

$$\text{Gross margin} = \frac{\text{Gross profit}}{S} \tag{5.3}$$

$$\text{Operating margin} = \frac{\text{Operating income}}{S} \tag{5.4}$$

$$\text{Net margin} = \frac{\text{NI}}{S} \tag{5.5}$$

Margin	What it measures (slides 41, 43)	What else moves it
Gross (5.3)	selling for more than the **direct** cost; the contribution to fixed costs and profit	—
Operating (5.4)	selling for more than direct **and indirect** cost; efficiency of operations	**market power** (mark-ups)
Net (5.5)	share of each sales dollar left to shareholders after all expenses, interest and taxes	**capital structure** (interest)

GM 2020 (slide 39 numbers, note 04 §6):

$$\text{gross} = \frac{13{,}672}{122{,}485} = 11.16\%, \quad \text{operating} = \frac{6{,}634}{122{,}485} = 5.42\%, \quad \text{net} = \frac{6{,}321}{122{,}485} = 5.16\%$$

Berk (pp. 35–36) adds that operating margins differ by **strategy** as well as efficiency: in 2010 Wal-Mart's was 5.87% and Nordstrom's 9.67%, with Wal-Mart's sales more than 45 times Nordstrom's.

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3 · C — economic earnings (slide 42)

"Does a positive net income imply that equityholders are properly compensated?" No. Owners earn an economic profit only if their return beats the **required** return on equity, which covers the time value of money and risk (slide 42). The accountant charges interest for debt and **nothing** for equity. Charging equity at its hurdle rate:

$$\text{Economic earnings} = \text{NI} - r_E\, E \tag{5.6}$$

Higgins, Polaris 2020 (pp. 27–28). NI = 414 million, E = 1,434 million, and an illustrative $r_E = 10\%$. Substituting into (5.6):

$$414 - 0.10 \times 1{,}434 = 414 - 143.4 = 270.6 \approx 271 \text{ million}$$

(Higgins's table rounds the charge to 143.) "The accountant pretends equity is free."

---

4 · C — receivable and payable days (slide 44)

$$\text{AR days} = \frac{AR}{S/365} \tag{5.7}$$

$$\text{AP days} = \frac{AP}{\text{purchases}/365} \tag{5.8}$$

AR days measure how long customers take to pay. **Berk, Global 2010 (pp. 36–37).** Average daily sales $186.7/365 = 0.5115$ million; substituting $AR = 18.5$ into (5.7):

$$\frac{18.5}{0.5115} = 36.2 \approx 36 \text{ days}$$

up from 27.5 days in 2009. Berk (p. 37) computes AP days with **cost of goods sold** in the denominator when purchases are not reported. **State which denominator you use.**

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5 · C — EBITDA and interest coverage (slide 45)

$$\text{EBITDA} = \text{EBIT} + \text{depreciation} + \text{amortization} \tag{5.9}$$

$$\text{Interest coverage} = \frac{\text{EBIT}}{\text{interest expense}} \tag{5.10}$$

Depreciation and amortization (the depreciation of intangibles) are not cash expenses, so EBITDA is "a better measure of the cash the firm obtains from its operations". Coverage measures the current ability to pay interest (slide 45). Berk (p. 37) notes that operating income or EBITDA are also used in the numerator of (5.10).

GM 2020 (EBIT from slide 39; D&A 12,815 from the cash-flow statement, slide 58):

$$\text{EBITDA} = 9{,}193 + 12{,}815 = 22{,}008$$

$$\text{coverage} = \frac{9{,}193}{1{,}098} = 8.37, \qquad \frac{\text{EBITDA}}{\text{interest}} = \frac{22{,}008}{1{,}098} = 20.04$$

Same leverage, different coverage: coverage depends on profitability as well as debt. A low coverage ratio does not bar borrowing if large cash flows are expected later, as for start-ups and fast growers (slide 45). The same logic applies to negative book equity (note 03 §6).

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6 · C — ROE, ROA, asset turnover and the DuPont identity (slides 46–48)

$$\text{ROE} = \frac{\text{NI}}{E} \tag{5.11}$$

$$\text{ROA} = \frac{\text{NI}}{A} \tag{5.12}$$

$$\text{Asset turnover} = \frac{S}{A} \tag{5.13}$$

ROE measures "the return the firm has earned on its past equity investments" and depends on capital structure. Turnover measures how efficiently assets generate sales (slide 46).

Deriving DuPont (Berk (2.17)–(2.18), p. 38). Multiply (5.11) by $S/S = 1$:

$$\text{ROE} = \frac{\text{NI}}{S} \cdot \frac{S}{E}$$

Multiply the second factor by $A/A = 1$:

$$\text{ROE} = \frac{\text{NI}}{S} \cdot \frac{S}{A} \cdot \frac{A}{E}$$

$$\boxed{\text{ROE} = \underbrace{\frac{\text{NI}}{S}}_{\text{net margin}} \times \underbrace{\frac{S}{A}}_{\text{turnover}} \times \underbrace{\frac{A}{E}}_{\text{financial leverage}}} \tag{5.14}$$

The first two factors are (5.12): $\frac{\text{NI}}{S} \cdot \frac{S}{A} = \frac{\text{NI}}{A}$, so

$$\text{ROE} = \text{ROA} \times \frac{A}{E} \qquad \text{and} \qquad \text{ROA} = \text{margin} \times \text{turnover} \tag{5.15}$$

Why is A/E "leverage"? Substitute $A = L + E$ from (1.1) (Higgins p. 39, fn. 1):

$$\frac{A}{E} = \frac{L+E}{E} = \frac{L}{E} + 1$$

which rises one for one with the liabilities-to-equity ratio (3.4).

Slide 48 (Higgins, 2013 data), each product recomputed:

Firm	Net margin	Turnover	Leverage	Product	Slide ROE
Google	21.6%	0.54	1.27	$0.216 \times 0.54 \times 1.27 = 14.81\%$	14.80%
JPMorgan	18.6%	0.04	11.44	$0.186 \times 0.04 \times 11.44 = 8.51\%$	8.50%
Hyundai Motor	10.3%	0.65	2.36	$0.103 \times 0.65 \times 2.36 = 15.80\%$	15.90%
Chevron	9.7%	0.87	1.7	$0.097 \times 0.87 \times 1.7 = 14.35\%$	14.40%

(Hyundai's 0.1-point gap is rounding in the printed inputs.)

> **Many roads lead to the same ROE.** JPMorgan reaches 8.5% with a turnover of 0.04, which only leverage of 11.44 can offset. Google gets 14.8% with almost no leverage. "Net profit margin and asset turnover tend to vary inversely: companies that add a lot of value to a product and thus can charge higher prices usually require lots of assets" (slide 48; Higgins 13e p. 40). Higgins's explanation of why ROEs cluster while their parts do not is **competition**: unusually high ROEs attract entry (p. 39).

Two more textbook cases. **Berk, Example 2.3 (p. 39):** Wal-Mart $3.6\% \times 2.4 \times 2.6 = 22.5\%$, Nordstrom $7.7\% \times 1.7 \times 2.4 = 31.4\%$; to match Nordstrom at its own turnover and leverage, Wal-Mart needs a margin of $31.4\%/(2.4 \times 2.6) = 31.4\%/6.24 = 5.0\%$. **Ross (p. 72):** Bank of America 2008, ROA 0.53% and ROE about 5.75%, so an equity multiplier of 10.85.

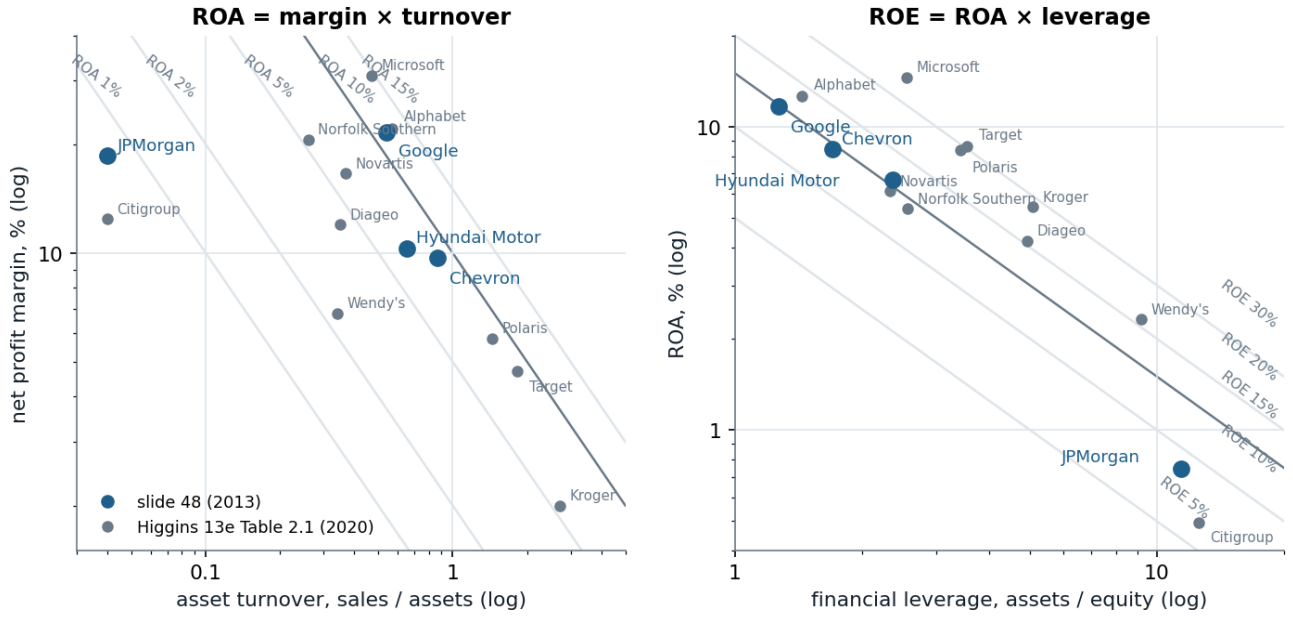


Figure 3: Two log-log panels. Left: net profit margin against asset turnover with straight iso-ROA lines for 1, 2, 5, 10 and 15 percent, the slide's four 2013 firms in blue and Higgins 13e's ten 2020 firms in grey. Right: ROA against financial leverage with iso-ROE lines for 5 to 30 percent; JPMorgan and Citigroup sit at leverage above 11 with ROA below 1 percent

Eq. (5.15) twice: the left panel multiplies margin by turnover, the right multiplies ROA by leverage; on log axes every iso-line is a straight line of slope −1. Data: slide 48 (2013) and Higgins 13e Table 2.1, p. 40 (2020).

On the left, Google and Alphabet sit near the 10% ROA line through high margins, Kroger and Target through high turnover, and the banks sit far below it. On the right, the banks climb back to ordinary ROEs on leverage alone. Their low ROA, high leverage position is the "bank equity" problem of L03 (block 2B.8).

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7 · C — ROE as a performance measure: three problems (slide 49)

"Is a company with a high ROE better…? Not necessarily" (slide 49). Higgins's three problems (pp. 54–57):

Problem	Statement	Worked case
Timing	backward-looking, one year	an equity issue to fund a project lowers ROE until the project pays (slide 49); start-up costs of a new product (Higgins p. 54)

Problem	Statement	Worked case
Risk	ignores the risk taken to earn it	Take-a-Risk: ROA 6% × leverage 5.0 = **30%**; Never-Dare: ROA 10% × leverage 2.0 = **20%**. Higgins prefers Never-Dare (pp. 54–55)
Value	return on **book**, not market, equity	Polaris: ROE 414/1,434 = 28.9%, but at the market price the return is 414/5,898 = 7.0% (p. 56)

Swapping in market equity gives the **earnings yield** (7.0% for Polaris), which has its own timing problem: "a bright future, a high stock price, and a low earnings yield go together" (Higgins p. 57).

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8 · C — ROIC and the exact leverage bridge (slide 50)

$$\text{ROIC} = \frac{\text{EBIT}(1-\tau)}{D+E} \tag{5.16}$$

This is also called return on net assets (RONA). The numerator is what the firm would earn after tax if all-equity financed. The denominator is every source of capital "on which a return must be earned". Accounts payable are excluded because "they carry no explicit cost" (Higgins p. 55). **Higgins, Polaris 2020 (p. 55):** EBIT 588, tax rate 107/521 = 0.2054, interest-bearing debt 142 + 1,294, equity 1,434:

$$\text{ROIC} = \frac{588 \times (1-0.2054)}{142+1{,}294+1{,}434} = \frac{467.2}{2{,}870} = 16.3\%$$

The two-firm example (slide 50, from Higgins). Identical firms, A levered, B all equity:

	Company A	Company B
Debt (10% interest)	900	0
Equity	100	1000
EBIT	120	120
Interest expense	90	0
Earnings before tax	30	120
Tax (40%)	12	48
Net income	18	72
ROE	18/100 = 18%	72/1000 = 7.2%
ROA	18/1000 = 1.8%	72/1000 = 7.2%
ROIC	120 × 0.6/1000 = 7.2%	120 × 0.6/1000 = 7.2%

ROE rewards A's leverage, ROA punishes it, and only ROIC sees two identical businesses. "ROIC thus reflects the company's fundamental earning power before it is confounded by differences in financing strategies" (Higgins p. 56).

Deriving the bridge. With interest $r_D D$, a proportional tax on pre-tax income, and no nonoperating income, net income is

$$\text{NI} = (\text{EBIT} - r_D D)(1 - \tau) \tag{5.17}$$

Expand the product:

$$\text{NI} = \text{EBIT}(1 - \tau) - r_D(1 - \tau)D \tag{5.18}$$

From (5.16), $\text{EBIT}(1 - \tau) = \text{ROIC}\,(D + E)$. Substitute into (5.18):

$$\text{NI} = (\text{ROIC}\,(D + E)) - r_D(1 - \tau)D \tag{5.19}$$

Divide by E and use (5.11):

$$\text{ROE} = \text{ROIC}\,\frac{D + E}{E} - r_D(1 - \tau)\frac{D}{E}$$

Split $\frac{D+E}{E} = 1 + \frac{D}{E}$:

$$\text{ROE} = \text{ROIC} + \text{ROIC}\,\frac{D}{E} - r_D(1 - \tau)\frac{D}{E}$$

Collect the D/E terms:

$$\boxed{\text{ROE} = \text{ROIC} + \frac{D}{E}\Big[\text{ROIC} - r_D(1 - \tau)\Big]} \tag{5.20}$$

Check on Company A. ROIC $= 0.072$, $r_D(1 - \tau) = 0.10 \times 0.60 = 0.060$, $D/E = 900/100 = 9$. Substituting into (5.20):

$$\text{ROE} = 0.072 + 9 \times (0.072 - 0.060) = 0.072 + 9 \times 0.012 = 0.072 + 0.108 = 0.180 \quad \checkmark$$

ROA under leverage. In the example $A = D + E$. Divide (5.19) by $D + E$:

$$\text{ROA} = \text{ROIC} - \frac{D}{D + E}\, r_D(1 - \tau) \tag{5.21}$$

For A: $0.072 - 0.9 \times 0.060 = 0.072 - 0.054 = 0.018$. ROA falls with **any** debt, whatever the spread.

The same firms in Higgins 13e (pp. 55–56) use $\tau = 25\%$: ROIC $= 120 \times 0.75/1000 = 9.0\%$, $r_D(1 - \tau) = 7.5\%$, and (5.20) gives

$$9.0\% + 9 \times (9.0\% - 7.5\%) = 9.0\% + 13.5\% = 22.5\%$$

which is Higgins's printed ROE for A (B: 9.0%).

Read (5.20). The assumptions are (5.17): interest proportional to book debt, taxes proportional to pre-tax income, with a refund when it is negative, and no nonoperating items. - Leverage raises ROE **iff** the spread ROIC $- r_D(1-\tau)$ is positive. At $r_D = 12\%$ with $\tau = 40\%$ the spread is zero and ROE stays at 7.2% for every D/E. At $r_D = 14\%$ leverage **lowers** ROE. - ROE is affine in D/E with slope equal to the spread: a straight line through $(0, \text{ROIC})$.

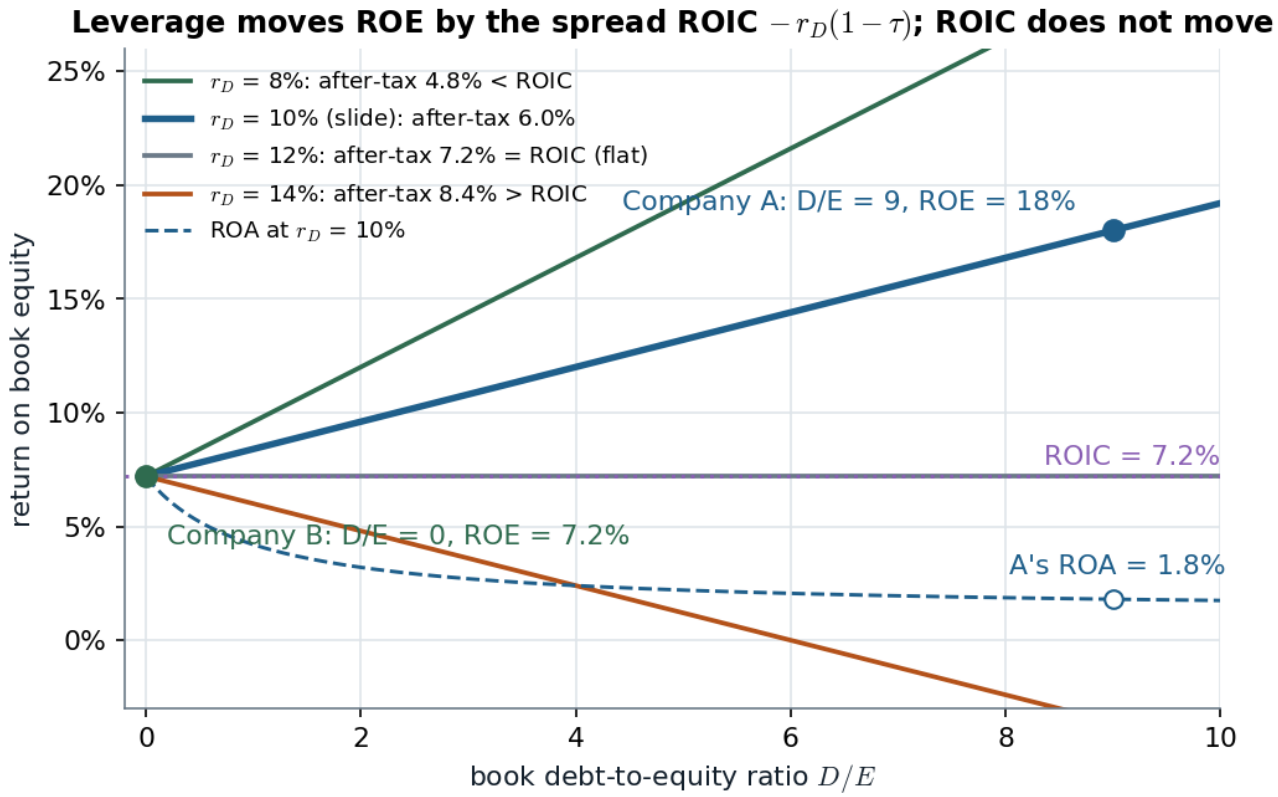


Figure 4: ROE against book D/E from 0 to 10 at a fixed ROIC of 7.2% for interest rates of 8, 10, 12 and 14 percent: four straight lines from the common point (0, 7.2%), rising for 8 and 10, flat for 12, falling for 14; Company A marked at D/E 9 with ROE 18% and Company B at D/E 0 with 7.2%; a dashed ROA curve falls to 1.8% at D/E 9

Eq. (5.20) at ROIC $= 7.2\%$, $\tau = 40\%$ *(slide 50), for four values of* r_D*; ROA from (5.21) at* $r_D = 10\%$.

Every line starts at Company B, since at $D/E = 0$ ROE is ROIC. The line through Company A has slope 1.2 points per unit of D/E, the spread. The flat line is the break-even rate $r_D = \text{ROIC}/(1-\tau) = 12\%$. ROA, dashed, falls with leverage on every line.

Interactive companion. Leverage and ROE (`companions/companion-leverage-roe.html`): move the debt share from 0 to 0.90 and the interest rate across 12% and watch the ROE line rotate around ROIC. The presets load Company A, Company B, the flat case, Higgins 13e's 25% tax and a negative spread.

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9 · C/G — "Should managers maximize financial leverage?" (slide 47)

DuPont (5.14) makes leverage look like a free lever. Three reasons it is not:

(i) Leverage scales risk as well as return. Rewrite (5.20) with ROIC collected:

$$\text{ROE} = \left(1 + \frac{D}{E}\right) \text{ROIC} - \frac{D}{E}\, r_D(1-\tau)$$

Hold D/E and r_D fixed and let ROIC vary from year to year. Then $\Delta\text{ROE} = (1 + D/E)\,\Delta\text{ROIC}$, so

$$\text{sd}(\text{ROE}) = \left(1 + \frac{D}{E}\right) \text{sd}(\text{ROIC}) \tag{5.22}$$

Company A's ROE is **ten times** as volatile as its ROIC. This is Higgins's risk problem (§7), made exact.

(ii) The required return rises too. A higher expected ROE on a riskier equity claim is compensation, not value. With frictionless markets the cost of equity rises one for one with this leverage effect, and firm value does not change (MM II, L02 block 2A.9). Higgins's footnote makes the same point from the investor's side: you can lever a safe firm's return yourself with personal borrowing (p. 55, fn. 3), which is **homemade leverage**.

(iii) Taxes and distress break the tie in opposite directions: interest is deductible and default is costly. The static trade-off is L03.

Berk adds a mechanical caveat (p. 38, fn. 7): raising leverage raises interest expense, which **lowers the net margin** in (5.14). The DuPont factors are not independent levers.

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10 · C — the price-earnings ratio (slide 51)

$$\frac{P}{E}\ \text{ratio} = \frac{E^{\text{mkt}}}{\text{NI}} = \frac{P}{\text{EPS}} \tag{5.23}$$

This is Berk's (2.19), p. 39. **Better growth prospects raise P/E, more risk lowers it**. P/E varies over time and across industries; a high P/E relative to industry peers *could* indicate overvaluation; and P/E is influenced by leverage (slide 51).

Berk, Global (p. 39): $36/2.0 = 18$ on totals, or $10/0.56 \approx 18$ per share. **Higgins, Polaris 2020 (p. 57):** $95.28/6.69 = 14.2$. When earnings are temporarily depressed P/E rises: Polaris's exceeded 45 in 2017 after earnings fell 19% (p. 57).

> **P/E depends on leverage because both its numerator and its denominator are equity measures.** The leverage-neutral alternatives use enterprise value (3.6) over EBIT or EBITDA (Berk pp. 39–40). How EPS and P/E respond to a debt-financed repurchase is PS2 Ex. 4, Topic 2A.

Links

- Prev / next: L01-04-income-statement · L01-06-cash-flows-fcf.
- The balance-sheet inputs: L01-02-balance-sheet-items, L01-03-book-vs-market-leverage.
- Forward: MM II and the cost of equity, L02 (blocks 2A.9–2A.10); bank capital and "ROE falls with equity", L03 (block 2B.8).
- Rules: rules/03 (MM benchmark).

- Practice: Lista R1 Ex. 5 (Higgins 2.3, p. 71), Ex. 7 (Berk 2.27, p. 56: DuPont), **Ex. 8 (Higgins 2.7, p. 73: ROE, ROA, ROIC, the identity (5.20))**, Ex. 12 (Berk 2.19, p. 54), Ex. 13 (Berk 2.23, p. 55: EPS and dilution); solutions PDF, checked by `Listas/lista-r1-accounting-capital-structure-solutions_check.py` (item 8 asserts (5.20)); quiz Topic 1 quiz tag `is`.
- Books: Higgins 13e Ch. 2 (pp. 38–40, 54–58); Berk 2e §2.5 (pp. 34–40); Ross *Fundamentos* Cap. 3 §3.4 (pp. 71–72).

1.8 · Profits are not cash: the cash-flow statement, FCF and FCFE

Claim. The cash-flow statement undoes the income statement's allocations. It adds back non-cash charges, nets out working-capital build-ups, and brings in capex and financing. **Free cash flow** is the cash the **assets** release after all investment, measured as if the firm were all-equity: $FCF = EBIT(1-\tau) + \text{Dep} - \text{Capex} - \Delta NWC$. It values the **business** (EV). **FCFE** subtracts after-tax interest and adds net borrowing; it values the **equity**. With taxes the two claims together receive $FCF + \tau \cdot \text{Int}$, not FCF. The gap is the tax shield, which later topics put in the discount rate (WACC) or price separately (APV).

Prev: L01-05-ratios-dupont-roic · Next: L01-07-formulas · Map: L01-00-map

Notation.

Symbol	Meaning	Range
NI, EBIT	net income, earnings before interest and taxes	real
Dep	depreciation and amortization of the period	≥ 0
$\Delta AR, \Delta AP, \Delta\text{Inv}$	increases in receivables, payables, inventories	real
ΔNWC	working-capital investment: increase in operating current assets minus increase in non-interest-bearing current liabilities	real
Capex	capital expenditures (purchases of PP&E)	≥ 0
Int, τ	interest expense; tax rate on pre-tax income	≥ 0; $[0, 1)$
ΔD	net borrowing: debt issued minus debt repaid	real
Div, ΔPIC	dividends (and repurchases); new paid-in capital	≥ 0; real
FCF, $FCFE$, CF_D	free cash flow to the firm, to equity, cash flow to debtholders	real

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1 · C — why profits are not cash (slides 52–53)

Net income does not measure the cash the firm generated (slide 52):

1. it contains **non-cash entries** (depreciation, amortization);
2. it **omits** some cash outlays (capital expenditures, spending on inventory).

So "it is possible that a profitable firm has not enough cash to meet its obligations". The statement of cash flows reports how much cash the firm generated and where it went, period by period, and investors care about the cash "they can extract from the business" (slide 53). **Berk's Global, 2010 (p. 44):** positive net income, yet **−0.4 million** of operating cash flow, "in large part because of the increase in accounts receivable".

2 · C — operating activities (slides 54–55)

Start from net income, add back non-cash charges, remove working-capital build-ups:

$$\text{CFO} = \text{NI} + \text{Dep} + \text{other non-cash items} - \Delta AR + \Delta AP - \Delta\text{Inv} \tag{6.1}$$

Adjustment	Sign	Why (slides 54–55)
Depreciation, amortization	+	deducted in NI, no cash leaves
Deferred taxes, accrued salaries and taxes	+	expensed, not yet paid
Increase in receivables	−	sales booked in NI, cash not received (eq. (4.1))
Increase in payables	+	purchases not yet paid
Increase in inventory	−	cash spent, not expensed (note 04 §3)

3 · C — investing and financing activities (slides 56–57)

Investing: capex is not in the income statement. Only its depreciation is, over time, and that was added back in (6.1). So the actual capex is **subtracted** here (slide 56). **Financing:** outflows to shareholders (dividends, repurchases) and changes in borrowing (new debt issued, principal repaid) (slide 57).

4 · C — worked example: GM, statement of cash flows 2020 (slide 58)

USD millions; every section subtotal recomputed:

Operating activities		Investing activities		Financing activities	
Net income	6,321	Expenditures for property	(5,300)	Net increase in short-term debt	277
Depreciation and amortization	12,815	AFS securities, acquisitions	(16,204)	Debt issuance (> 3 months)	78,527
Accounts receivable	(1,341)	AFS securities, liquidation	11,941	Debt payments (> 3 months)	(72,663)
Accounts payable	42	Other adjustments	(12,263)	Stock issuance	492
Inventories	(104)			Dividends paid	(669)

Operating activities		Investing activities		Financing activities	
Other adjustments	(1,063)			Other financing activities	(412)
Net cash, operating	**16,670**	**Net cash, investing**	**(21,826)**	**Net cash, financing**	**5,552**

$$\text{CFO} = 6{,}321 + 12{,}815 - 1{,}341 + 42 - 104 - 1{,}063 = 16{,}670$$

$$\text{CFI} = -5{,}300 - 16{,}204 + 11{,}941 - 12{,}263 = -21{,}826$$

$$\text{CFF} = 277 + 78{,}527 - 72{,}663 + 492 - 669 - 412 = 5{,}552$$

Add the foreign-exchange adjustment of -222:

$$\Delta\text{Cash} = 16{,}670 - 21{,}826 + 5{,}552 - 222 = 174$$

$$\text{Cash}_{\text{end}} = 22{,}943 + 174 = 23{,}117 \quad \checkmark$$

Two readings. (i) D&A (12,815) is twice net income. GM's operating cash dwarfs its profit because the fleet and plants are depreciated, not expensed. (ii) The AFS lines are **cash management**, buying and selling marketable securities, and not investment in operations. Only "Expenditures for property" (5,300) is capex in (6.6).

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5 · C — the cash identity and the measures of cash flow (slide 59)

Deriving slide 59's identity. Split assets into cash and non-cash assets NCA, and book equity into paid-in capital and retained earnings:

$$\text{Cash} + NCA = L + PIC + RE \tag{6.2}$$

Take differences between two dates:

$$\Delta\text{Cash} = \Delta L + \Delta PIC + \Delta RE - \Delta NCA \tag{6.3}$$

With no other equity changes, retained earnings grow by net income less dividends (eq. (4.9)):

$$\Delta RE = \text{NI} - \text{Div} \tag{6.4}$$

Substitute (6.4) into (6.3):

$$\Delta\text{Cash} = \Delta L + \Delta PIC + (\text{NI} - \text{Div}) - \Delta NCA$$

$$\boxed{\Delta\text{cash} = \text{NI} - \Delta\text{noncash assets} + \Delta\text{liabilities} + \Delta\text{paid-in capital} - \text{dividends}} \tag{6.5}$$

Read it. Every line of the cash-flow statement is a line of (6.5): ΔNCA collects receivables, inventory and net PP&E (capex minus depreciation, hence the depreciation add-back); ΔL collects payables and borrowing. The statement is the balance-sheet identity (1.1) in differences.

The slide's three measures, from narrow to broad (slides 59–60; Higgins pp. 20–21):

Measure	Definition	Higgins, Polaris 2020
Net cash flow ("cash earnings")	NI + non-cash items	640 million
Cash flow from operating activities	net cash flow ± changes in current assets and liabilities, i.e. − change in NWC	changes in current items added 306 million
Free cash flow	cash available to owners and creditors after funding all worthwhile investment	"essentially cash flow from operating activities less capital expenditures" (p. 21)

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6 · C — free cash flow to the firm and to equity (slides 60–61)

FCF (unlevered free cash flow), slide 60:

$$\boxed{FCF = \text{EBIT}(1-\tau) + \text{Dep} - \text{Capex} - \Delta NWC} \tag{6.6}$$

Term	Why (slide 60)
$\text{EBIT}(1-\tau)$	the income earned "regardless of how the business is financed": taxes charged on EBIT, as if there were no interest
+Dep	non-cash, as in (6.1)
−Capex	the investment the income statement left out (§3)
$-\Delta NWC$	increase in current assets needed by operations, **minus** the increase in **non-interest-bearing** current liabilities (payables, accruals; short-term debt is financing)

FCFE (levered free cash flow), slide 61: FCF minus after-tax interest, plus net borrowing:

$$\boxed{FCFE = FCF - \text{Int}(1-\tau) + \Delta D} \tag{6.7}$$

Link to net income. With no nonoperating items, $\text{NI} = (\text{EBIT} - \text{Int})(1-\tau)$ (eq. (5.17)). Subtract $\text{Int}(1-\tau)$ from (6.6):

$$FCF - \text{Int}(1-\tau) = \text{EBIT}(1-\tau) - \text{Int}(1-\tau) + \text{Dep} - \text{Capex} - \Delta NWC$$

The first two terms are NI:

$$FCF - \text{Int}(1-\tau) = \text{NI} + \text{Dep} - \text{Capex} - \Delta NWC \tag{6.8}$$

Substitute (6.8) into (6.7):

$$FCFE = \text{NI} + \text{Dep} - \Delta NWC - \text{Capex} + \Delta D \tag{6.9}$$

The first three terms are CFO (6.1) without other items. So FCFE is "essentially cash flow from operating activities minus capital expenditures" plus net debt issuance "if the firm is changing its level of debt" (slide 61).

Worked example: GM 2020 (slides 39, 58). Inputs: EBIT 9,193; $\tau = 1{,}774/8{,}095 = 0.21915$ (note 04 §6); D&A 12,815; capex 5,300; and from the operating section

$$\Delta NWC = 1{,}341 - 42 + 104 = 1{,}403$$

(receivables up 1,341, payables up 42, inventories up 104). Substituting into (6.6):

$$FCF = 9{,}193 \times (1 - 0.21915) + 12{,}815 - 5{,}300 - 1{,}403$$
$$FCF = 7{,}178.4 + 12{,}815 - 5{,}300 - 1{,}403 = 13{,}290.4$$

Net borrowing from the financing section:

$$\Delta D = 277 + 78{,}527 - 72{,}663 = 6{,}141$$

After-tax interest: $1{,}098 \times (1 - 0.21915) = 857.4$. Substituting into (6.7):

$$FCFE = 13{,}290.4 - 857.4 + 6{,}141 = 18{,}574.0$$

Check with (6.8): $13{,}290.4 - 857.4 = 12{,}433.0$ and $6{,}321 + 12{,}815 - 5{,}300 - 1{,}403 = 12{,}433$. The two sides agree exactly because τ is the effective rate T/EBT. GM's own CFO minus capex is $16{,}670 - 5{,}300 = 11{,}370$; the 1,063 difference is "other adjustments", which (6.8) leaves out.

Eqs. (6.6)–(6.7) on slides 39 and 58; τ *is GM's effective rate 1,774/8,095.*

The tallest bar is the depreciation add-back, larger than NOPAT itself. FCFE exceeds FCF because GM borrowed 6.1 bn net in 2020, much of it at GM Financial. **For a firm with a captive finance arm, "net borrowing" funds the loan book, an operating asset.** FCFE overstates what the owners could take out, and the operating/financing split behind (6.6) is blurred (note 04 §6).

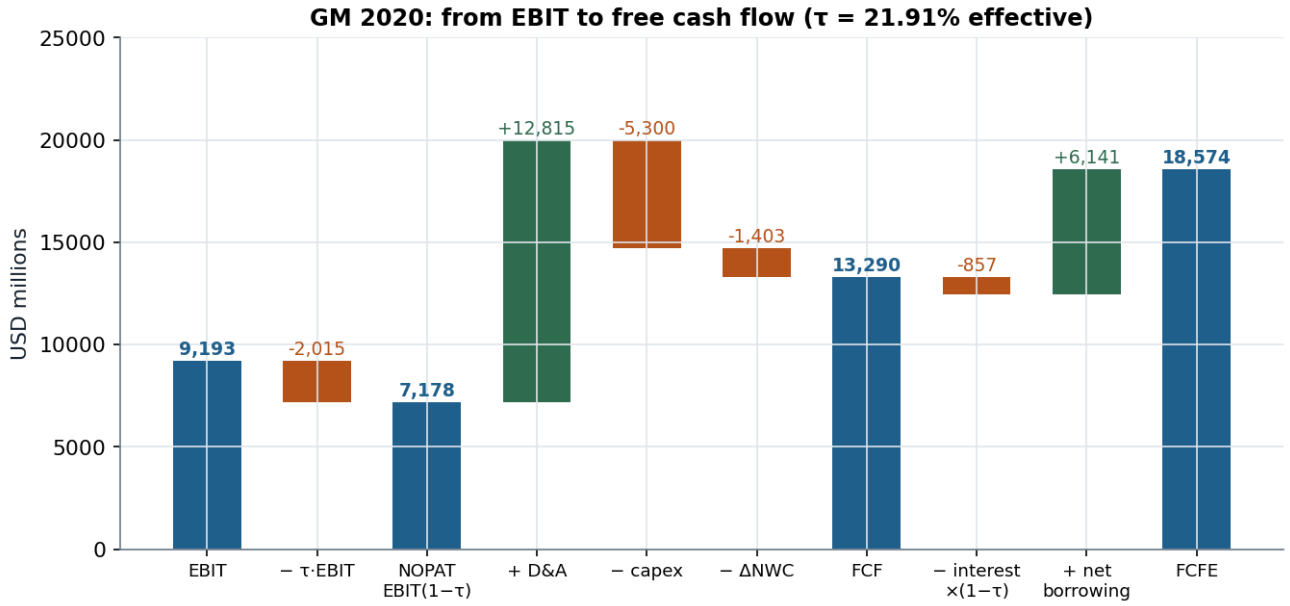


Figure 5: Waterfall for GM 2020 in USD millions: EBIT 9,193, minus tax on EBIT 2,015, gives NOPAT 7,178; plus D&A 12,815, minus capex 5,300, minus the NWC increase 1,403, gives FCF 13,290; minus after-tax interest 857, plus net borrowing 6,141, gives FCFE 18,574

7 · C/G — who receives FCF: the tax claimant reappears (slides 13, 60–61)

Cash flow to debtholders is interest paid minus net new borrowing:

$$CF_D = \text{Int} - \Delta D \tag{6.10}$$

Add (6.7) and (6.10):

$$FCFE + CF_D = (FCF - \text{Int}(1-\tau) + \Delta D) + (\text{Int} - \Delta D)$$

$$FCFE + CF_D = FCF + \tau\,\text{Int} \tag{6.11}$$

The two claims receive more than FCF, by the interest tax shield $\tau \cdot$ Int. FCF (6.6) charges taxes on EBIT, but the firm actually pays taxes on EBIT − Int. Slide 13's $FCF_A = FCF_L + FCF_E$ (eq. (1.6)) is therefore exact when $\tau = 0$, or when FCF_A means the actual cash flow from assets, measured with actual taxes:

$$CF_A^{\text{actual}} = \text{EBIT} + \text{Dep} - T - \text{Capex} - \Delta NWC \tag{6.12}$$

With $T = \tau(\text{EBIT} - \text{Int})$, (6.12) equals $FCF + \tau\,\text{Int}$, and so, by (6.11):

$$CF_A^{\text{actual}} = CF_D + FCFE \tag{6.13}$$

Ross's Brasil S/A (pp. 38–41) is (6.12)–(6.13). EBIT 694, depreciation 65, taxes 212, net capital spending 130, ΔNWC 330, interest 70, net borrowing 46 (note 01 §6). Then

$$\tau = \frac{212}{694 - 70} = \frac{212}{624} = 0.33974$$

Substituting into (6.6):

$$FCF = 694 \times (1 - 0.33974) + 65 - 130 - 330 = 458.22 - 395 = 63.22$$

The tax shield is $\tau \cdot \text{Int} = 0.33974 \times 70 = 23.78$, and

$$FCF + \tau\,\text{Int} = 63.22 + 23.78 = 87.00 = CF_A^{\text{actual}} = 24 + 63 \quad \checkmark$$

FCFE by (6.7): $63.22 - 70 \times 0.66026 + 46 = 63.22 - 46.22 + 46 = 63.00$, Ross's cash flow to shareholders.

> **Why the slides use unlevered FCF anyway.** (6.6) is a property of the business alone and does not change when financing changes. The tax benefit of debt is then priced separately: in the discount rate (after-tax WACC) or as its own present value (APV). That is Topic 2B, L03 (blocks 2B.1–2B.4).

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8 · C — what each cash flow values, and where FCF goes (slides 61–62)

Forecast of	Discounted to	Value
FCF (6.6)	the firm's (unlevered or WACC) rate	**enterprise value** (3.6)
FCFE (6.7)	the cost of equity	**market equity value**

Slide 62's tree of uses:

```
FCF to the firm ─┬─ after-tax payment to debtholders
                 └─ FCF to equity ─┬─ retain ─┬─ invest in new projects
                                   │          └─ increase cash reserves
                                   └─ pay out ─┬─ share repurchase
                                               └─ dividend
```

> **Two open questions the tree leaves for later.** Does the split between retaining and paying out, or between dividends and buybacks, change value? With fixed investment, no: payout irrelevance, L03 block 2B.9. What do managers do with free cash they are not forced to pay out? That is the agency and liquidity side of the syllabus (rules/04, rules/05), and the "increase cash reserves" branch is slide 17's rise in corporate cash (note 02 §3).

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9 · C — the problem set (PS1 Ex. 1)

PS1 Exercise 1 is written on this note. Part (a), net income versus FCF, uses §1 and the term-by-term table of §6. Part (b), deferred taxes in the FCF formula, starts from slide 54, which lists deferred taxes among the non-cash add-backs of (6.1), and from the EBIT$(1 - \tau)$ term of (6.6). The statement is in `Listas/PS1.pdf`. It is not reproduced here.

Links

- Prev / next: L01-05-ratios-dupont-roic · L01-07-formulas.
- The identity behind (6.13): L01-01-balance-sheet-view §6; depreciation and accruals: L01-04-income-statement §2–3; working capital: L01-02-balance-sheet-items §5.
- Forward: FCF as the valuation input (incremental, real/nominal, terminal value), L02 block 2A.3; why unlevered FCF ignores the tax shield, L03 blocks 2B.1, 2B.4.
- Rules: rules/02.
- Practice: PS1 Ex. 1(a)–(b) (`Listas/PS1.pdf`); Lista R1 Ex. 3 (Berk 2.31, p. 57), Ex. 9 (Ross 2.14, p. 48), Ex. 10 (Higgins 1.11, p. 34), Ex. 11 (Ross 2.21, p. 49), Ex. 15 (Ross 2.25–2.26, pp. 50–51); quiz Topic 1 quiz tag `cf`.
- Books: Higgins 13e Ch. 1 (pp. 18–21); Berk 2e §2.6 (pp. 42–45); Ross *Fundamentos* Cap. 2 §2.4 (pp. 37–41), Tabela 2.5 (p. 41).

Topic 1 · Formula sheet: every definition, every number, the lecture's content mapped

Prev: L01-06-cash-flows-fcf · Next: L01-08-pitfalls · Map: L01-00-map

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0 · The exam specification, mapped

The slides state one goal (slide 6): "introduction to basic financial statements", the "key to measuring and understanding relevant aspects of firms". The content list is the block map of lectures-index (Topic 1), item by item:

Block	Content item (slides)	Note
1.1	Asset pricing, corporate finance, banking (4–5)	L01-01-balance-sheet-view §1
1.2	$A = L + E$ in book **and** market value (7–8)	L01-01-balance-sheet-view §3, §6
1.2	Recognition: tangible vs intangible; historical cost vs market value; SVB (9–10)	L01-01-balance-sheet-view §4
1.2	Goods **and promises**; liabilities senior, equity junior and residual (11–12)	L01-01-balance-sheet-view §5
1.2	$FCF_A = FCF_L + FCF_E \Rightarrow A = L + E$ (13)	L01-01-balance-sheet-view §6; L01-06-cash-flows-fcf §7
1.2	Cash flow–production cycle (14)	L01-01-balance-sheet-view §7
1.3	Current vs non-current assets and liabilities; goodwill (15–16, 18–19)	L01-02-balance-sheet-items §1–2, §4
1.3	Depreciation is not a cash expense; book value = cost − accumulated depreciation (16)	L01-02-balance-sheet-items §2
1.3	Rise in corporate cash (17)	L01-02-balance-sheet-items §3
1.3	Net working capital; GPA (20–21)	L01-02-balance-sheet-items §5–6
1.4	Market cap; forward- vs backward-looking; off-balance-sheet items (22–23)	L01-03-book-vs-market-leverage §1–2
1.4	Starbucks; intangibles; market-to-book (24–27)	L01-03-book-vs-market-leverage §3–5
1.5	D/E book vs market; net debt; EV; current and quick ratios; covenants (28–31)	L01-03-book-vs-market-leverage §6–8
1.6	Accrual and matching; depreciation spreads capex (32–35)	L01-04-income-statement §2–3

Block	Content item (slides)	Note
1.6	Gross profit, opex, operating leverage, EBIT, net income (36–37)	L01-04-income-statement §4
1.6	Statement of equity; GM income statement (38–39)	L01-04-income-statement §5–6
1.7	EPS, diluted EPS; margins; economic earnings (40–43)	L01-05-ratios-dupont-roic §1–3
1.7	AR/AP days; EBITDA; coverage (44–45)	L01-05-ratios-dupont-roic §4–5
1.7	ROE, ROA, turnover, **DuPont**; "maximize leverage?" (46–48)	L01-05-ratios-dupont-roic §6, §9
1.7	ROE pitfalls; **ROIC**; P/E (49–51)	L01-05-ratios-dupont-roic §7–8, §10
1.8	Profits ≠ cash; operating, investing, financing sections; GM (52–58)	L01-06-cash-flows-fcf §1–4
1.8	Net cash flow, CFO, Δcash identity (59)	L01-06-cash-flows-fcf §5
1.8	**FCF**, **FCFE**, what each values, uses of FCF (60–62)	L01-06-cash-flows-fcf §6–8

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1 · Every numbered equation

Eq.	Statement	Derived / used in
(1.1)	$A = L + E$	L01-01-balance-sheet-view §3 (Berk (2.1), p. 26)
(1.2)	$\Delta A = \Delta L + \Delta E$	§3
(1.3)	creditors $\min(X, F)$, owners $\max(X - F, 0)$	§5
(1.4)	$\min(X, F) + \max(X - F, 0) = X$	§5
(1.5)	$FCF_A = FCF_L + FCF_E \Rightarrow A = L + E$ (slide 13)	§6
(1.6)	the split holds date by date, state by state	§6
(1.7)	linear pricing $\mathrm{PV}(Y + Z) = \mathrm{PV}(Y) + \mathrm{PV}(Z)$	§6
(1.8)	$A^{\mathrm{mkt}} = L^{\mathrm{mkt}} + E^{\mathrm{mkt}}$	§6
(2.1)	goodwill = price paid − fair value of identifiable net assets	L01-02-balance-sheet-items §2
(2.2)	straight line $\mathrm{Dep} = (K_0 - S_v)/n$	§2
(2.3)	$BV_k = K_0 - \sum_{j \le k} \mathrm{Dep}_j$	§2
(2.4)	$NWC = CA - CL$	§5 (Berk (2.2), p. 28)
(3.1)	$E^{\mathrm{book}} = A - L$	L01-03-book-vs-market-leverage §1

Eq.	Statement	Derived / used in
(3.2)	$E^{\text{mkt}} = P \times N$	§1
(3.3)	market-to-book $= E^{\text{mkt}}/E^{\text{book}}$	§5 (Berk (2.3), p. 30)
(3.4)	D/E (slide: L/E; Berk (2.4): debt/equity)	§6
(3.5)	net debt $= D - C$	§6
(3.6)	$EV = E^{\text{mkt}} + D - C$	§7 (Berk (2.5), p. 31)
(3.7)	current ratio CA/CL	§8
(3.8)	quick ratio $(CA - \text{Inv})/CL$	§8
(4.1)	cash collected $= S - \Delta AR$	L01-04-income-statement §2
(4.2)–(4.5)	gross profit, operating income, EBIT, NI	§4
(4.6)	$\text{OI} = (1-v)S - F$	§4
(4.7)	$\Delta\text{OI} = (1-v)\Delta S$	§4
(4.8)	elasticity of OI to sales $= 1 + F/\text{OI}$	§4
(4.9)	$E_{\text{end}} = E_{\text{begin}} + \text{NI} - \text{Div} \pm \text{other}$	§5
(5.1)–(5.2)	EPS, diluted EPS	L01-05-ratios-dupont-roic §1
(5.3)–(5.5)	gross, operating, net margin	§2
(5.6)	economic earnings $= \text{NI} - r_E E$	§3
(5.7)–(5.8)	AR days, AP days	§4
(5.9)–(5.10)	EBITDA, interest coverage	§5
(5.11)–(5.13)	ROE, ROA, asset turnover	§6
(5.14)	**DuPont** ROE = margin × turnover × leverage	§6 (Berk (2.18), p. 38)
(5.15)	ROE = ROA $\times$ A/E; ROA = margin × turnover	§6
(5.16)	$\text{ROIC} = \text{EBIT}(1-\tau)/(D+E)$	§8
(5.17)–(5.19)	NI in terms of EBIT and ROIC	§8
(5.20)	ROE $=$ ROIC $+ \frac{D}{E}[\text{ROIC} - r_D(1-\tau)]$	§8
(5.21)	$\text{ROA} = \text{ROIC} - \frac{D}{D+E} r_D(1-\tau)$	§8
(5.22)	$\text{sd(ROE)} = (1 + D/E)\,\text{sd(ROIC)}$	§9
(5.23)	$P/E = E^{\text{mkt}}/\text{NI} = P/\text{EPS}$	§10 (Berk (2.19), p. 39)
(6.1)	$\text{CFO} = \text{NI} + \text{Dep} + \text{other} - \Delta AR + \Delta AP - \Delta\text{Inv}$	L01-06-cash-flows-fcf §2
(6.2)–(6.4)	balance sheet split, differences, ΔRE	§5
(6.5)	$\Delta\text{cash} = \text{NI} - \Delta NCA + \Delta L + \Delta PIC - \text{Div}$ (slide 59)	§5
(6.6)	$FCF = \text{EBIT}(1-\tau) + \textbf{Dep} - \textbf{Capex} - \Delta NWC$	§6
(6.7)	$FCFE = FCF - \textbf{Int}(1-\tau) + \Delta D$	§6
(6.8)	$FCF - \text{Int}(1-\tau) = \text{NI} + \text{Dep} - \text{Capex} - \Delta NWC$	§6
(6.9)	$FCFE = \text{NI} + \text{Dep} - \Delta NWC - \text{Capex} + \Delta D$	§6

Eq.	Statement	Derived / used in
(6.10)	$CF_D = \text{Int} - \Delta D$	§7
(6.11)	$FCFE + CF_D = FCF + \tau\,\text{Int}$	§7
(6.12)	actual cash flow from assets, with actual taxes	§7
(6.13)	$CF_A^{\text{actual}} = CF_D + FCFE$	§7

1.1 · Definitions without an equation number

Term	Definition (slide)
Current asset / liability	converted to cash / due within one year (15, 18)
Accrual principle	revenue at sale, expense at purchase, not at payment (32)
Matching principle	revenues and expenses in the period of the transaction (32)
Operating leverage	high share of fixed costs → vulnerable to sales declines (36)
Dilution	growth in shares from options and convertibles (40)
Growth / value stock	high / low market-to-book (27)
Net cash flow (cash earnings)	NI + non-cash items (59; Higgins p. 21)
Working-capital investment	increase in operating current assets − increase in non-interest-bearing current liabilities (60)

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2 · Every number and worked case

Case	Numbers	Note
Plant on credit (slide 9)	$\Delta A = \Delta L = +1$ bn	01 §3
Ross Brasil S/A 2009 (pp. 38–41)	OCF 547, NCS 130, ΔNWC 330 → CFA 87 = 24 + 63; unlevered FCF 63.22, $\tau \cdot$ Int 23.78	01 §6, 06 §7
Intangibles ≈ 90% of S&P 500 EV (slide 9)	—	01 §4
Higgins facility (p. 13)	50 m, 10 years → 5 m/yr; $BV_3 = 35$	02 §2, 04 §3
Corporate cash (slide 17)	< 1 tn (1994) → ≈ 4 tn (2017); cash/GDP index 100 → ≈ 160 (read off the chart)	02 §3
Berk Global NWC (p. 28)	57.0 − 34.7 = 22.3	02 §5
GPA 2020 (slide 21)	A 53,295; L 36,488; E 16,807; **NWC −842**; payables 61.8% of CL, 21.4% of A; current 0.954, quick 0.601; book L/E 2.171	02 §6, 03 §6, §8

Case	Numbers	Note
Berk Global (pp. 29–31)	market cap 36 vs book 22.2; D 118.7; book D/E 5.3, market 3.3; EV 131.5	03 §1, §6, §7
Polaris (Higgins p. 25)	market cap 5,898 vs book 1,434 (4.1×)	03 §1
Starbucks 2020 (slides 24–25)	**book E −7,904**; market cap 122.6 bn; debt 15,917; net debt 10,889; **EV 133,489** (142,512 with leases); market L/E 0.309, D/E 0.130; current 1.060, quick 0.873	03 §3–8
Amazon June 2005 (Berk p. 29)	book E −64 m, market > 15 bn	03 §3
Intangible share (slide 26)	17.1% (1975) → 84.0% (2018)	03 §4
Market-to-book (Berk p. 30)	AmeriServ 0.52, Citigroup 0.82, large U.S. ≈ 3, Amazon > 11	03 §5
Heinz (Berk Ex. 2.2, p. 32)	cap 14.59 bn, book 1.83, D 4.83, EV 18.858 bn	03 §7
Ross Exemplo 3.1 (p. 62)	4/2 → 3/1 = 3; 2/4 → 1/3	03 §8
Polaris receivables (Higgins pp. 12–13)	revenue 7,108, ΔAR 67 → cash 7,041	04 §2
GM 2020 P&L (slide 39)	sales 122,485; gross 13,672; OI 6,634; EBIT 9,193; pre-tax 8,095; NI 6,321; **τ = 21.915%**; DOL 2.061	04 §4, §6
Berk Global retained earnings (p. 44)	2 − 1 = 1; payout 50%	04 §5
GM margins and coverage	gross 11.16%, operating 5.42%, net 5.16%; EBITDA 22,008; coverage 8.37 (20.04 on EBITDA)	05 §2, §5
Polaris economic earnings (Higgins pp. 27–28)	414 − 143 = 271	05 §3
Berk Global AR days (p. 37)	18.5 / 0.5115 = 36 days (27.5 in 2009)	05 §4
DuPont 2013 (slide 48)	Google 14.8, JPMorgan 8.5, Hyundai 15.9, Chevron 14.4 (%)	05 §6
Berk Ex. 2.3 (p. 39)	Wal-Mart 22.5%, Nordstrom 31.4%, required margin 5.0%	05 §6
Higgins risk problem (p. 54)	6% × 5.0 = 30% vs 10% × 2.0 = 20%	05 §7
Higgins value problem (p. 56)	ROE 28.9% vs earnings yield 7.0%	05 §7
Polaris ROIC (Higgins p. 55)	467.2 / 2,870 = 16.3%	05 §8
Two firms (slide 50)	A: ROE 18%, ROA 1.8%, ROIC 7.2%; B: 7.2% three times; flat at $r_D = 12\%$	05 §8

Case	Numbers	Note
Same firms, Higgins 13e (pp. 55–56)	$\tau = 25\%$: A's ROE 22.5%, ROIC 9.0%	05 §8
P/E	Berk Global 18 (p. 39); Polaris 14.2 (Higgins p. 57)	05 §10
Berk Global CFO (p. 44)	positive NI, CFO −0.4 m	06 §1
GM 2020 cash flows (slide 58)	CFO 16,670; CFI −21,826; CFF 5,552; FX −222; Δcash 174; 22,943 → 23,117	06 §4
GM FCF bridge	NOPAT 7,178.4; ΔNWC 1,403; **FCF 13,290.4**; after-tax interest 857.4; net borrowing 6,141; **FCFE 18,574.0**	06 §6

Topic 1 · Pitfalls and a True/False self-test

Prev: L01-07-formulas · Next: L01-09-python · Map: L01-00-map

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Part I — the confusions this material produces

1 · Negative book equity is not insolvency

Counterexample: Starbucks, book equity −7,904 m and market cap 122.6 bn (slides 24–25). Book equity is the residual (3.1) of historical costs and cumulative payouts (4.9). Solvency is about whether the assets' **market** value covers the claims. → L01-03-book-vs-market-leverage §3.

2 · Profit is not cash

Counterexample: Berk's Global, positive net income and −0.4 m of operating cash flow (p. 44), because receivables grew. Cash collected is $S - \Delta AR$ (4.1). → L01-06-cash-flows-fcf §1.

3 · "Depreciation is a source of cash"

Adding depreciation back in (6.1) only **undoes** a non-cash charge. **Minimal counterexample:** with $\tau = 0$, doubling depreciation lowers NI by the same amount it raises the add-back, so CFO does not move. Depreciation's only cash effect runs through taxes: each unit of depreciation saves τ in tax, which is why $\text{EBIT}(1-\tau) + \text{Dep}$ in (6.6) rises with Dep when $\tau > 0$.

4 · Which "D/E"?

The slide divides **total liabilities** by equity (3.4). Berk divides **interest-bearing debt** (p. 30). Starbucks' market ratio is 0.309 on the first and 0.130 on the second. Net debt subtracts cash as well (3.5). State the definition, and say book or market. → L01-03-book-vs-market-leverage §6.

5 · EV is not "market cap + total liabilities"

(3.6) adds **debt** and subtracts **cash**. Operating liabilities such as payables are part of the business, so they stay out. Leases are a convention choice: Starbucks' EV is 133.5 bn without them and 142.5 bn with them. → §7.

6 · A higher ROE is not better performance

Three failures (Higgins pp. 54–57). **Timing:** an equity-funded project lowers ROE before it pays. **Risk:** Take-a-Risk's 30% is worse than Never-Dare's 20%. **Value:** ROE is on **book** equity (Polaris 28.9% vs a 7.0% earnings yield). → L01-05-ratios-dupont-roic §7.

7 · Leverage raises ROE only with a positive spread, and always raises its risk

By (5.20), ROE rises with D/E iff $\text{ROIC} > r_D(1-\tau)$. **Counterexample:** slide 50's firm at $r_D = 16\%$ has ROE −14.4%. Even with a positive spread, (5.22) multiplies ROE's volatility by $1 + D/E$. → §8–9.

8 · ROA "measures operations"

ROA has net income, which is after interest, in the numerator, so it **falls with any debt** (5.21): Company A shows 1.8% against B's 7.2% with identical operations. Only ROIC (5.16) is financing-neutral. → §8.

9 · DuPont's factors are independent levers

Raising leverage raises interest and **lowers the net margin** in (5.14) (Berk p. 38, fn. 7). The identity is exact, but it says nothing about causation. → §6, §9.

10 · FCF uses actual taxes

(6.6) charges tax on **EBIT**, as if there were no debt. Actual taxes are lower by $\tau \cdot \text{Int}$, so the claims together receive $FCF + \tau\,\text{Int}$ (6.11). Mixing in actual taxes and then also using an after-tax WACC counts the tax shield **twice**. → L01-06-cash-flows-fcf §7.

11 · Interest belongs in FCF

Interest is a payment **to a claimant**, so it is excluded from FCF, as are dividends. Subtracting it gives a cash flow to equity, and FCFE also needs net borrowing (6.7). Discounting FCF − interest at a firm-wide rate mixes the two valuations of §8.

12 · Every current item is in ΔNWC

Short-term **debt** and current maturities are financing. Working-capital investment nets only **non-interest-bearing** current liabilities (slide 60; Berk p. 28). Securities bought for cash management, such as GM's AFS lines, are not capex either. → §4, §6.

13 · A better current ratio means more liquidity

Paying current liabilities with cash **raises** a ratio above 1 and **lowers** one below 1 (Ross Exemplo 3.1, p. 62). Negative NWC can also be a business model (GPA, supplier finance). → L01-03-book-vs-market-leverage §8, L01-02-balance-sheet-items §6.

14 · Positive net income means owners are compensated

Net income charges nothing for equity. Polaris's 414 m becomes 271 m of economic earnings at a 10% cost of equity (5.6). → L01-05-ratios-dupont-roic §3.

15 · A high P/E means overvalued

P/E rises with growth, falls with risk, and moves with leverage (slide 51). It signals overvaluation only **relative to comparable peers**, and even then only "could". Depressed earnings inflate it (Polaris above 45 in 2017). → §10.

Part II — True/False self-test (original items)

Scoring as in `macro-1`*:* $\max\{0, \textit{correct} - \textit{wrong}/2\}$*; blanks do not count.*

1. A firm whose total liabilities exceed its total book assets must have negative book equity. **T.** Book equity is defined as the residual $A - L$ (3.1). If $L > A$ the residual is negative by construction, as in Starbucks (126% of assets in liabilities). Nothing says the firm's market value is negative.

2. $A = L + E$ holds in market value only if the firm has no debt. **F.** (1.8) follows from linear pricing and the exhaustive split (1.6), for **any** split of the cash flow. Debt changes who receives each state's cash, not the sum, as long as the split leaves the cash flow itself unchanged.

3. Buying a machine for cash changes total assets but not book equity. **F.** It changes the **composition** of assets (cash down, PP&E up by the same amount). Total assets, liabilities and equity are all unchanged. Only later depreciation reduces equity, through net income.

4. Straight-line depreciation lowers the firm's cash in every year of the asset's life. **F.** The cash leaves once, at purchase (slide 34). Depreciation allocates that outlay over time (note 02 §2). Its only cash effect is the tax it saves when $\tau > 0$.

5. A firm can report positive net income and negative cash flow from operations in the same year. **T.** Receivable or inventory build-ups are subtracted in (6.1). Berk's Global did exactly this (p. 44).

6. Under the accrual principle, a customer paying an old invoice raises revenue in the period of payment. **F.** Slide 33, row 2: cash flow yes, revenue no. The revenue was booked at the sale.

7. With all of cost of sales variable and all SG&A fixed, GM's 2020 operating income would fall about 20.6% after a 10% fall in sales. **T.** By (4.8) the elasticity is $1 + 7{,}038/6{,}634 = 2.061$, and $2.061 \times 10\% = 20.6\%$; the direct computation in note 04 §4 gives -20.61%.

8. Market-to-book above 1 means the firm's accountants overstated its liabilities. **F.** It means the market values the equity above its historical-cost residual. Forward-looking value and off-balance-sheet assets (slide 23) explain it, with no accounting error required.

9. For a firm with large cash balances, net debt is a better leverage measure than gross debt. **T.** Slide 29: cash can repay debt on short notice, so it acts as negative debt.

10. Enterprise value equals market capitalization plus total liabilities. **F.** (3.6) adds **debt** and subtracts **cash**. Operating liabilities such as payables and accruals stay out because they belong to the business being valued.

11. If two firms have the same ROE, they have the same net profit margin. **F.** DuPont (5.14) has three factors. Google (14.8%) and Chevron (14.4%) reach similar ROEs with margins of 21.6% and 9.7% (slide 48).

12. Holding ROIC, r_D and τ fixed, ROE is a linear function of the book debt-equity ratio. **T.** (5.20) is affine in D/E with intercept ROIC and slope $\text{ROIC} - r_D(1-\tau)$.

13. Raising leverage always raises ROE. **F.** Only if the spread is positive. At $r_D(1-\tau) > \text{ROIC}$ leverage lowers ROE; slide 50's firm at $r_D = 16\%$ gets -14.4%.

14. In slide 50's example, ROA rates Company A's operations below Company B's even though they are identical. **T.** ROA is 1.8% vs 7.2% because A's net income is after interest (5.21). Only ROIC (7.2% for both) is financing-neutral.

15. With $D/E = 9$, a one-point fall in ROIC lowers ROE by nine points. **F.** By ten: by (5.20), $\partial\text{ROE}/\partial\text{ROIC} = 1 + D/E = 10$ (5.22).

16. Positive accounting net income implies that shareholders earned at least their required return. **F.** Net income charges no cost for equity (slide 42). Economic earnings (5.6) can be negative while NI is

positive.

17. Free cash flow to the firm subtracts interest expense because interest is a cash outflow. **F.** Interest is a payment to a claimant. FCF (6.6) is measured before any payment to claimants and taxes EBIT as if the firm were all-equity.

18. FCFE can exceed FCF. **T.** By (6.7), when net borrowing exceeds after-tax interest. GM 2020: FCF 13,290 and FCFE 18,574 (note 06 §6).

19. With corporate taxes, the cash flow to debtholders plus FCFE equals unlevered FCF exactly. **F.** They sum to $FCF + \tau\,\text{Int}$ (6.11). Ross's Brasil S/A: $24 + 63 = 87 = 63.22 + 23.78$.

20. An increase in accounts payable raises cash flow from operations, holding everything else fixed. **T.** Purchases expensed (or put in inventory) but not yet paid: the sign is + in (6.1).

21. Diluted EPS can never exceed basic EPS when options and convertibles are counted in the denominator as on slide 40. **T.** With the slide's definition (5.2), the numerator is the same NI and the denominator is at least N, so for positive NI the diluted figure is weakly smaller. (Accounting standards refine this with the convertible's interest and anti-dilution rules, which are not on the slides.)

22. A higher P/E than industry peers proves the stock is overvalued. **F.** Slide 51 says it "could indicate" overvaluation. Higher growth, lower risk, or a different capital structure can all justify it.

Links

- Every pitfall points to its note; equations in L01-07-formulas.
- More practice in the same style: Topic 1 quiz (tags `bs`, `is`, `cf`); Lista R1 Ex. 5 (Higgins 2.3, p. 71, true/false on ratios); grading style rules/00.

Topic 1 · Python — every routine, checked against the slides' and books' numbers

> **Claim.** The accounting review is arithmetic on identities, so its reference implementation is a set of small functions, one per numbered equation, each followed by `asserts` on the source's own printed numbers. The block below runs in about a second. It re-derives every worked case of notes 01–06, then **runs the existing Lista R1 checker** instead of re-implementing its 15 exercises.

Prev: L01-08-pitfalls · Next: — · Map: L01-00-map

Run. Save the block as a file and run it from the repo root with `python -I <file>` (standard library only; `cwd` must be the repo root, for the Lista R1 checker path). The verification run of 2026-10-24 printed:

```
L01 python: all asserts pass; Lista R1 checker: PASS 163 checks
```

What is reused, and why the rest is new. `Listas/lista-r1-accounting-capital-structure-solutions_check.py` is a flat script of checks on the list's 15 exercises (Higgins 1.4 … Ross 2.25–2.26), with no functions to import. So it is **executed** (last section), not copied. Its item 8 already asserts the leverage bridge (5.20) on Higgins 2.7's numbers. The lecture's own examples (GPA, Starbucks, GM, slide 48, slide 50) are in no existing script, so the functions below are new. Figures use their own copies of the same expressions in figures/make_figures.py, and the companion is checked by companions/check_companions.py.

Section of the code	Equations	Note
balance sheet	(1.1), (2.4), (3.4)–(3.8)	L01-02-balance-sheet-items, L01-03-book-vs-market-leverage
income statement	(4.2)–(4.5), (4.8)	L01-04-income-statement
ratios, DuPont, ROIC	(5.6)–(5.23)	L01-05-ratios-dupont-roic
cash flows	(6.1), (6.5)–(6.13)	L01-06-cash-flows-fcf

The reference implementation

```
"""L01 reference implementation: every routine the accounting review needs, checked against the
slides' and books' own numbers. Run from the repo root: python -I <this file>."""
import pathlib
import subprocess
import sys

sys.stdout.reconfigure(encoding="utf-8")


def close(a, b, tol):
    """True if |a - b| <= tol; tol is half a unit of the last printed decimal."""
    return abs(a - b) <= tol


# ---- balance sheet: identity (1.1), NWC (2.4), ratios (3.3)-(3.8) ------------------------
```

```
def nwc(ca, cl):
    """Net working capital, eq. (2.4)."""
    return ca - cl


def current_ratio(ca, cl):
    """Eq. (3.7)."""
    return ca / cl


def quick_ratio(ca, inv, cl):
    """Eq. (3.8): current assets other than inventory over current liabilities."""
    return (ca - inv) / cl


def enterprise_value(mcap, debt, cash):
    """Eq. (3.6): market equity + debt - cash."""
    return mcap + debt - cash


GPA = dict(ca=17641, nca=35654, ta=53295, cash=8711, inv=6536, ap=11424, cl=18483,
 ↪  ncl=18005,
           tl=36488, e=16807)                                        # slide 21, BRL m
SBUX = dict(ca=8358, nca=21611, ta=29968, cash=5028, inv=1472, cl=7884, ncl=29989,
 ↪  tl=37872,
            e=-7904, debt=493 + 750 + 14674, mcap=122600)       # slides 24-25, USD
 ↪  m

assert GPA["ca"] + GPA["nca"] == GPA["ta"] == GPA["tl"] + GPA["e"]       # (1.1)
assert SBUX["tl"] + SBUX["e"] == SBUX["ta"]                              # (1.1),
 ↪  residual -7,904
assert nwc(GPA["ca"], GPA["cl"]) == -842 and nwc(SBUX["ca"], SBUX["cl"]) == 474
assert close(current_ratio(GPA["ca"], GPA["cl"]), 0.954, 5e-4)
assert close(quick_ratio(GPA["ca"], GPA["inv"], GPA["cl"]), 0.601, 5e-4)
assert close(current_ratio(SBUX["ca"], SBUX["cl"]), 1.060, 5e-4)
assert close(quick_ratio(SBUX["ca"], SBUX["inv"], SBUX["cl"]), 0.873, 5e-4)
assert close(GPA["ap"] / GPA["cl"], 0.6181, 5e-5)                        # suppliers'
 ↪  share, note 02 §6
assert SBUX["debt"] == 15917
assert enterprise_value(SBUX["mcap"], SBUX["debt"], SBUX["cash"]) == 133489
assert close(enterprise_value(36, 118.7, 23.2), 131.5, 5e-2)            # Berk
 ↪  Global, p. 31
assert close(SBUX["tl"] / SBUX["mcap"], 0.309, 5e-4)                     # market
 ↪  D/E, total liabilities
assert close(118.7 / 22.2, 5.3, 5e-2) and close(118.7 / 36, 3.3, 5e-2)  # Berk p. 31,
 ↪  book vs market D/E

# ---- income statement (4.2)-(4.5), operating leverage (4.8)
 ↪  ------------------------------
def income_statement(sales, cogs, opex, noi, interest, taxes):
```

```
    """Eqs. (4.2)-(4.5): returns gross profit, operating income, EBIT, pre-tax
     ↪ income, NI."""
    gp = sales - cogs
    oi = gp - opex
    ebit = oi + noi
    ebt = ebit - interest
    return dict(gp=gp, oi=oi, ebit=ebit, ebt=ebt, ni=ebt - taxes)


def dol(fixed, oi):
    """Eq. (4.8): elasticity of operating income to sales, 1 + F/OI (v and F
     ↪ constant)."""
    return 1 + fixed / oi


gm = income_statement(108673 + 13812, 97539 + 11274, 7038, 2559, 1098, 1774)   #
 ↪ slide 39
assert (gm["gp"], gm["oi"], gm["ebit"], gm["ebt"], gm["ni"]) == (13672, 6634, 9193,
 ↪ 8095, 6321)
tau_gm = 1774 / gm["ebt"]
assert close(tau_gm, 0.21915, 5e-6)
assert close(dol(7038, 6634), 2.061, 5e-4)
assert close(((0.9 * 13672 - 7038) - 6634) / 6634, -0.2061, 5e-5)       # 10% sales
 ↪ fall, note 04 §4

# ---- ratios (5.3)-(5.10), DuPont (5.14), ROIC bridge (5.16)-(5.22)
 ↪ -------------------------
S = 122485
assert close(gm["gp"] / S, 0.1116, 5e-5) and close(gm["oi"] / S, 0.0542, 5e-5)
assert close(gm["ni"] / S, 0.0516, 5e-5)
assert close(gm["ebit"] / 1098, 8.37, 5e-3)                                    # coverage
 ↪ (5.10)
assert gm["ebit"] + 12815 == 22008                                             # EBITDA
 ↪ (5.9)
assert close(414 - 0.10 * 1434, 270.6, 5e-2)                                   # economic
 ↪ earnings (5.6)
assert close(18.5 / (186.7 / 365), 36.2, 5e-2)                                 # AR days
 ↪ (5.7), Berk p. 37


def dupont(margin, turnover, leverage):
    """Eq. (5.14): ROE = margin x turnover x leverage."""
    return margin * turnover * leverage


for name, (roe, m, t, lev) in {"Google": (14.8, 21.6, 0.54, 1.27), "JPMorgan": (8.5,
 ↪ 18.6, 0.04, 11.44),
                               "Hyundai Motor": (15.9, 10.3, 0.65, 2.36), "Chevron":
                                ↪ (14.4, 9.7, 0.87, 1.7)}.items():
```

```
    assert close(dupont(m, t, lev), roe, 0.15), name                    # slide 48:
     ↪ inputs are rounded
assert close(dupont(3.6, 2.4, 2.6), 22.5, 5e-2) and close(dupont(7.7, 1.7, 2.4),
 ↪ 31.4, 5e-2)  # Berk Ex. 2.3


def returns(ebit, debt, equity, rd, tau):
    """Statement route: (5.17), then ROE (5.11), ROA (5.12) with assets = D + E, ROIC
     ↪ (5.16)."""
    ni = (ebit - rd * debt) * (1 - tau)
    return dict(ni=ni, roe=ni / equity, roa=ni / (debt + equity), roic=ebit * (1 -
     ↪ tau) / (debt + equity))


def roe_bridge(roic, de, rd, tau):
    """Eq. (5.20): ROE = ROIC + (D/E)(ROIC - r_D(1 - tau))."""
    return roic + de * (roic - rd * (1 - tau))


for tau, roe_a, roe_b in ((0.40, 0.18, 0.072), (0.25, 0.225, 0.09)):     # slide 50;
 ↪ Higgins 13e pp. 55-56
    a, b = returns(120, 900, 100, 0.10, tau), returns(120, 0, 1000, 0.10, tau)
    assert close(a["roe"], roe_a, 1e-12) and close(b["roe"], roe_b, 1e-12)
    assert close(a["roe"], roe_bridge(a["roic"], 9, 0.10, tau), 1e-12)   # both
     ↪ routes agree
    assert close(a["roic"], b["roic"], 1e-12)                            # ROIC is
     ↪ financing-neutral
    assert close(a["roa"], a["roic"] - 0.9 * 0.10 * (1 - tau), 1e-12)    # (5.21)
assert close(returns(120, 900, 100, 0.12, 0.40)["roe"], 0.072, 1e-12)    # zero
 ↪ spread: flat
assert close(returns(120, 900, 100, 0.16, 0.40)["roe"], -0.144, 1e-12)   # negative
 ↪ spread
d1 = returns(130, 900, 100, 0.10, 0.40)["roe"] - returns(120, 900, 100, 0.10,
 ↪ 0.40)["roe"]
d0 = 130 * 0.6 / 1000 - 120 * 0.6 / 1000
assert close(d1 / d0, 10.0, 1e-9)                                         # (5.22):
 ↪ swing 1 + D/E
assert close(588 * (1 - 107 / 521) / (142 + 1294 + 1434), 0.163, 5e-4)   # Polaris
 ↪ ROIC, p. 55
assert close(95.28 / 6.69, 14.2, 5e-2)                                    # Polaris
 ↪ P/E (5.23), p. 57

# ---- cash flows (6.1), (6.5)-(6.13)
 ↪ ----------------------------------------------------------
cfo = 6321 + 12815 - 1341 + 42 - 104 - 1063                               # (6.1),
 ↪ slide 58
cfi = -5300 - 16204 + 11941 - 12263
cff = 277 + 78527 - 72663 + 492 - 669 - 412
assert (cfo, cfi, cff) == (16670, -21826, 5552)
assert cfo + cfi + cff - 222 == 174 and 22943 + 174 == 23117
```

```python
def fcf(ebit, tau, dep, capex, dnwc):
    """Eq. (6.6): unlevered free cash flow."""
    return ebit * (1 - tau) + dep - capex - dnwc


def fcfe(fcf_, interest, tau, net_borrowing):
    """Eq. (6.7): FCF minus after-tax interest plus net borrowing."""
    return fcf_ - interest * (1 - tau) + net_borrowing


dnwc = 1341 - 42 + 104
f = fcf(9193, tau_gm, 12815, 5300, dnwc)
fe = fcfe(f, 1098, tau_gm, 277 + 78527 - 72663)
assert dnwc == 1403 and close(f, 13290.4, 0.05) and close(fe, 18574.0, 0.05)
assert close(f - 1098 * (1 - tau_gm), 6321 + 12815 - 5300 - dnwc, 1e-9)  # (6.8)

# Ross, Brasil S/A (pp. 38-41): (6.11)-(6.13)
tau_r = 212 / (694 - 70)
f_r = fcf(694, tau_r, 65, 130, 330)
cf_d, cf_e = 70 - 46, 103 - 40
assert close(f_r, 63.22, 5e-3) and close(tau_r * 70, 23.78, 5e-3)
assert close(f_r + tau_r * 70, 87, 1e-9) and cf_d + cf_e == 87          # (6.11),
 ↪  (6.13)
assert close(fcfe(f_r, 70, tau_r, 46), cf_e, 1e-9)                      # (6.7)
 ↪  reproduces Ross's 63

# ---- reuse: the list's own checker
 ↪  ------------------------------------------------------------
root = pathlib.Path.cwd()
chk = root / "Listas" / "lista-r1-accounting-capital-structure-solutions_check.py"
out = subprocess.run([sys.executable, "-I", str(chk)], capture_output=True,
 ↪  text=True, check=True).stdout
assert out.startswith("PASS"), out
print("L01 python: all asserts pass;", "Lista R1 checker:", out.strip())
```

Links

- The notes each section checks: L01-01-balance-sheet-view (Ross identity), L01-02-balance-sheet-items, L01-03-book-vs-market-leverage, L01-04-income-statement, L01-05-ratios-dupont-roic, L01-06-cash-flows-fcf; every equation listed in L01-07-formulas.
- Reused checker: `Listas/lista-r1-accounting-capital-structure-solutions_check.py` (solutions: PDF).
- Code conventions: rules/lang-python.